Theory of Mathematics

Volume III: Measure, Fourier Analysis, Distributions and PDE

Contents

Part I

Measure and Integration

Chapter 1

Sigma-Algebras and Measures

Measure theory begins by specifying which sets are stable under countable operations and which size functions are countably additive. The resulting continuity properties are the set-theoretic engine behind integration.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

1.1 Sigma-algebras

A sigma-algebra contains the empty set and is closed under complements and countable unions; De Morgan's laws give closure under countable intersections and differences.

1.2 Generated sigma-algebras

The sigma-algebra generated by a family is the intersection of all sigma-algebras containing it, giving the smallest measurable structure compatible with the generators.

1.3 Borel sets

The Borel sigma-algebra is generated by open sets. On the real line, open intervals with rational endpoints already generate all Borel sets.

1.4 Measures

A measure is a nonnegative countably additive set function with value zero at the empty set. Countable additivity forces monotonicity and subadditivity.

1.5 Finite and sigma-finite measures

Finite measures have finite total mass; sigma-finite spaces are countable unions of finite-measure pieces and are the natural setting for product theory.

1.6 Continuity from below

If measurable sets increase to a union, their measures increase to the measure of that union.

1.7 Continuity from above

For decreasing measurable sets with finite measure at the first stage, measures decrease to the measure of the intersection.

1.8 Completion and null sets

A complete measure space contains every subset of every measurable null set, so null-set modifications remain measurable.

1.9 Core structural results

Theorem 1.1 (Continuity from below)**.** *If $E_n \uparrow E$, then $\mu(E_n) \uparrow \mu(E)$.*

Proof. Disjointify the increase by writing E as E_1 together with the successive differences $E_n \setminus E_{n-1}$, then use countable additivity. □

Theorem 1.2 (Continuity from above)**.** *If $E_n \downarrow E$ and $\mu(E_1) < \infty$, then $\mu(E_n) \downarrow \mu(E)$.*

Proof. Apply continuity from below to $E_1 \setminus E_n \uparrow E_1 \setminus E$ and subtract from the finite quantity $\mu(E_1)$. □

Theorem 1.3 (Countable subadditivity)**.** *For measurable E_n, $\mu(\cup_n E_n) \leq \sum_n \mu(E_n)$.*

Proof. Replace the family by disjoint pieces $F_n = E_n \setminus \cup_{k<n} E_k$, use countable additivity, and compare $\mu(F_n) \leq \mu(E_n)$. □

1.10 Worked examples

Example 1.4 (Sigma-algebras diagnostic)**.** Give a rigorous worked analysis of Sigma-algebras in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A sigma-algebra contains the empty set and is closed under complements and countable unions; De Morgan's laws give closure under countable intersections and differences. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 1.5 (Generated sigma-algebras diagnostic)**.** Give a rigorous worked analysis of Generated sigma-algebras in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The sigma-algebra generated by a family is the intersection of all sigma-algebras containing it, giving the smallest measurable structure compatible with the generators. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 1.6 (Borel sets diagnostic)**.** Give a rigorous worked analysis of Borel sets in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The Borel sigma-algebra is generated by open sets. On the real line, open intervals with rational endpoints already generate all Borel sets. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 1.7 (Measures diagnostic)**.** Give a rigorous worked analysis of Measures in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A measure is a nonnegative countably additive set function with value zero at the empty set. Countable additivity forces monotonicity and subadditivity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

1.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 1.8 (Sigma-algebras diagnostic)**.** Give a rigorous worked analysis of Sigma-algebras in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A sigma-algebra contains the empty set and is closed under complements and countable unions; De Morgan's laws give closure under countable intersections and differences. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 1.9 (Generated sigma-algebras diagnostic)**.** Give a rigorous worked analysis of Generated sigma-algebras in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The sigma-algebra generated by a family is the intersection of all sigma-algebras containing it, giving the smallest measurable structure compatible with the generators. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 1.10 (Borel sets diagnostic)**.** Give a rigorous worked analysis of Borel sets in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The Borel sigma-algebra is generated by open sets. On the real line, open intervals with rational endpoints already generate all Borel sets. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 1.11 (Measures diagnostic)**.** Give a rigorous worked analysis of Measures in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A measure is a nonnegative countably additive set function with value zero at the empty set. Countable additivity forces monotonicity and subadditivity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 1.12 (Finite and sigma-finite measures diagnostic)**.** Give a rigorous worked analysis of Finite and sigma-finite measures in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Finite measures have finite total mass; sigma-finite spaces are countable unions of finite-measure pieces and are the natural setting for product theory. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 1.13 (Continuity from below diagnostic)**.** Give a rigorous worked analysis of Continuity from below in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. If measurable sets increase to a union, their measures increase to the measure of that union. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 1.14 (Continuity from above diagnostic)**.** Give a rigorous worked analysis of Continuity from above in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For decreasing measurable sets with finite measure at the first stage, measures decrease to the measure of the intersection. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 1.15 (Completion and null sets diagnostic)**.** Give a rigorous worked analysis of Completion and null sets in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A complete measure space contains every subset of every measurable null set, so null-set modifications remain measurable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 1.16 (Continuity from below proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $E_n \uparrow E$, then $\mu(E_n) \uparrow \mu(E)$.

Solution. Disjointify the increase by writing E as E_1 together with the successive differences $E_n \setminus E_{n-1}$, then use countable additivity. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 1.17 (Continuity from above proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $E_n \downarrow E$ and $\mu(E_1) < \infty$, then $\mu(E_n) \downarrow \mu(E)$.

Solution. Apply continuity from below to $E_1 \setminus E_n \uparrow E_1 \setminus E$ and subtract from the finite quantity $\mu(E_1)$. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 1.18 (Countable subadditivity proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For measurable E_n, $\mu(\cup_n E_n) \leq \sum_n \mu(E_n)$.

Solution. Replace the family by disjoint pieces $F_n = E_n \setminus \cup_{k<n} E_k$, use countable additivity, and compare $\mu(F_n) \leq \mu(E_n)$. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 1.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Measure theory begins by specifying which sets are stable under countable operations and which size functions are countably additive. The resulting continuity properties are the set-theoretic engine behind integration. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

1.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 1.20. State and apply the central principle behind Sigma-algebras. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A sigma-algebra contains the empty set and is closed under complements and countable unions; De Morgan's laws give closure under countable intersections and differences. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 1.21. State and apply the central principle behind Generated sigma-algebras. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The sigma-algebra generated by a family is the intersection of all sigma-algebras containing it, giving the smallest measurable structure compatible with the generators. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 1.22. State and apply the central principle behind Borel sets. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The Borel sigma-algebra is generated by open sets. On the real line, open intervals with rational endpoints already generate all Borel sets. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 1.23. State and apply the central principle behind Measures. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A measure is a nonnegative countably additive set function with value zero at the empty set. Countable additivity forces monotonicity and subadditivity. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 1.24. State and apply the central principle behind Finite and sigma-finite measures. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Finite measures have finite total mass; sigma-finite spaces are countable unions of finite-measure pieces and are the natural setting for product theory. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 1.25. State and apply the central principle behind Continuity from below. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. If measurable sets increase to a union, their measures increase to the measure of that union. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 1.26. State and apply the central principle behind Continuity from above. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. For decreasing measurable sets with finite measure at the first stage, measures decrease to the measure of the intersection. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 1.27. State and apply the central principle behind Completion and null sets. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A complete measure space contains every subset of every measurable null set, so null-set modifications remain measurable. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 2

Measurable Functions

Measurable functions are those whose inverse images respect the measurable structure. Their closure under algebra, limits, and monotone simple approximation makes them the natural domain of Lebesgue integration.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

2.1 Inverse-image definition

A real-valued function is measurable when inverse images of Borel sets are measurable.

2.2 Ray criteria

It is enough to test sets of the form $\{f < a\}$ or $\{f > a\}$ for all real a because such rays generate the Borel sigma-algebra.

2.3 Algebraic operations

Sums, products, absolute values, maxima, and minima of measurable real-valued functions remain measurable.

2.4 Pointwise limits

Countable suprema, infima, limsup, liminf, and pointwise limits of measurable functions are measurable.

2.5 Simple functions

A simple measurable function has finite range and measurable level sets; it is the finite-data model for integration.

2.6 Approximation from below

Every nonnegative measurable function is the increasing pointwise limit of nonnegative simple functions obtained by dyadic quantization and truncation.

2.7 Almost-everywhere equivalence

On complete measure spaces, changing a measurable function on a null set preserves measurability.

2.8 Composition

Composing a measurable map with a continuous or Borel measurable scalar map preserves measurability.

2.9 Core structural results

Theorem 2.1 (Pointwise limit theorem)**.** *A pointwise limit of measurable real-valued functions is measurable.*

Proof. Represent the limit using limsup and liminf, each built from countable suprema and infima of measurable functions. ☐

Theorem 2.2 (Algebra closure)**.** *If f and g are measurable, then $f+g$ and fg are measurable.*

Proof. The map $x \mapsto (f(x), g(x))$ is measurable into $\mathbb{R}^2$, and addition and multiplication are continuous Borel maps. ☐

Theorem 2.3 (Simple approximation)**.** *If $f \geq 0$ is measurable, there exist simple $\phi_n \uparrow f$.*

Proof. Truncate at height n and replace values by the lower endpoints of dyadic intervals of width 2^{-n}; measurability of the level sets gives simple functions and the mesh tends to zero. ☐

2.10 Worked examples

Example 2.4 (Inverse-image definition diagnostic)**.** Give a rigorous worked analysis of Inverse-image definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A real-valued function is measurable when inverse images of Borel sets are measurable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 2.5 (Ray criteria diagnostic)**.** Give a rigorous worked analysis of Ray criteria in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. It is enough to test sets of the form $\{f < a\}$ or $\{f > a\}$ for all real a because such rays generate the Borel sigma-algebra. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 2.6 (Algebraic operations diagnostic)**.** Give a rigorous worked analysis of Algebraic operations in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Sums, products, absolute values, maxima, and minima of measurable real-valued functions remain measurable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 2.7 (Pointwise limits diagnostic)**.** Give a rigorous worked analysis of Pointwise limits in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Countable suprema, infima, limsup, liminf, and pointwise limits of measurable functions are measurable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

2.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 2.8 (Inverse-image definition diagnostic)**.** Give a rigorous worked analysis of Inverse-image definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A real-valued function is measurable when inverse images of Borel sets are measurable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 2.9 (Ray criteria diagnostic)**.** Give a rigorous worked analysis of Ray criteria in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. It is enough to test sets of the form $\{f < a\}$ or $\{f > a\}$ for all real a because such rays generate the Borel sigma-algebra. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 2.10 (Algebraic operations diagnostic)**.** Give a rigorous worked analysis of Algebraic operations in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Sums, products, absolute values, maxima, and minima of measurable real-valued functions remain measurable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 2.11 (Pointwise limits diagnostic)**.** Give a rigorous worked analysis of Pointwise limits in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Countable suprema, infima, limsup, liminf, and pointwise limits of measurable functions are measurable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 2.12 (Simple functions diagnostic)**.** Give a rigorous worked analysis of Simple functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A simple measurable function has finite range and measurable level sets; it is the finite-data model for integration. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 2.13 (Approximation from below diagnostic)**.** Give a rigorous worked analysis of Approximation from below in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Every nonnegative measurable function is the increasing pointwise limit of nonnegative simple functions obtained by dyadic quantization and truncation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 2.14 (Almost-everywhere equivalence diagnostic)**.** Give a rigorous worked analysis of Almost-everywhere equivalence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. On complete measure spaces, changing a measurable function on a null set preserves measurability. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 2.15 (Composition diagnostic)**.** Give a rigorous worked analysis of Composition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Composing a measurable map with a continuous or Borel measurable scalar map preserves measurability. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 2.16 (Pointwise limit theorem proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: A pointwise limit of measurable real-valued functions is measurable.

Solution. Represent the limit using limsup and liminf, each built from countable suprema and infima of measurable functions. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 2.17 (Algebra closure proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If f and g are measurable, then $f + g$ and fg are measurable.

Solution. The map $x \mapsto (f(x), g(x))$ is measurable into $\mathbb{R}^2$, and addition and multiplication are continuous Borel maps. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 2.18 (Simple approximation proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f \geq 0$ is measurable, there exist simple $\phi_n \uparrow f$.

Solution. Truncate at height n and replace values by the lower endpoints of dyadic intervals of width 2^{-n}; measurability of the level sets gives simple functions and the mesh tends to zero. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 2.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Measurable functions are those whose inverse images respect the measurable structure. Their closure under algebra, limits, and monotone simple approximation makes them the natural domain of Lebesgue integration. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. ☐

2.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 2.20. State and apply the central principle behind Inverse-image definition. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. A real-valued function is measurable when inverse images of Borel sets are measurable. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 2.21. State and apply the central principle behind Ray criteria. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. It is enough to test sets of the form $\{f < a\}$ or $\{f > a\}$ for all real a because such rays generate the Borel sigma-algebra. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 2.22. State and apply the central principle behind Algebraic operations. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Sums, products, absolute values, maxima, and minima of measurable real-valued functions remain measurable. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 2.23. State and apply the central principle behind Pointwise limits. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Countable suprema, infima, limsup, liminf, and pointwise limits of measurable functions are measurable. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 2.24. State and apply the central principle behind Simple functions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A simple measurable function has finite range and measurable level sets; it is the finite-data model for integration. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 2.25. State and apply the central principle behind Approximation from below. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Every nonnegative measurable function is the increasing pointwise limit of nonnegative simple functions obtained by dyadic quantization and truncation. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 2.26. State and apply the central principle behind Almost-everywhere equivalence. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. On complete measure spaces, changing a measurable function on a null set preserves measurability. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 2.27. State and apply the central principle behind Composition. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Composing a measurable map with a continuous or Borel measurable scalar map preserves measurability. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 3

The Lebesgue Integral

The Lebesgue integral is constructed by integrating simple functions and taking monotone suprema. It is insensitive to null-set changes, linear on integrable functions, and controlled by the integral of the absolute value.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

3.1 Simple-function integral

For a nonnegative simple function $\phi = \sum a_k 1_{E_k}$ on disjoint measurable level sets, define $\int \phi = \sum a_k \mu(E_k)$.

3.2 Nonnegative functions

For measurable $f \geq 0$, define the integral as the supremum of simple-function integrals over $0 \leq \phi \leq f$.

3.3 Positive and negative parts

Write $f = f^+ - f^-$ with $f^+, f^- \geq 0$ and $|f| = f^+ + f^-$.

3.4 Integrable functions

A measurable real-valued function is integrable when $\int |f| < \infty$, which rules out the indeterminate expression $\infty - \infty$.

3.5 Monotonicity

If $0 \leq f \leq g$, then $\int f \leq \int g$ because every simple lower approximation to f is also one for g.

3.6 Linearity

Integrable functions form a vector space and the integral is linear.

3.7 Null-set invariance

If $f = g$ almost everywhere and one is integrable, then both represent the same integral.

3.8 Integrals over sets

Define $\int_E f = \int f 1_E$, making restriction to measurable subsets part of the same formalism.

3.9 Core structural results

Theorem 3.1 (Absolute integral inequality)**.** *For integrable f, $|\int f| \le \int |f|$.*

Proof. Use $-|f| \le f \le |f|$ and monotonicity of the integral. □

Theorem 3.2 (Zero integral criterion)**.** *If $f \ge 0$ and $\int f = 0$, then $f = 0$ almost everywhere.*

Proof. For $E_n = \{f \ge 1/n\}$, the inequality $\int f \ge (1/n)\mu(E_n)$ forces every E_n null; their union is $\{f > 0\}$. □

Theorem 3.3 (Linearity)**.** *If f, g are integrable and $a, b \in \mathbb{R}$, then $\int(af + bg) = a\int f + b\int g$.*

Proof. Prove first for nonnegative simple functions on a common refinement, pass to nonnegative measurable functions by monotone approximation, then use positive/negative parts. □

3.10 Worked examples

Example 3.4 (Simple-function integral diagnostic)**.** Give a rigorous worked analysis of Simple-function integral in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For a nonnegative simple function $\phi = \sum a_k 1_{E_k}$ on disjoint measurable level sets, define $\int \phi = \sum a_k \mu(E_k)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 3.5 (Nonnegative functions diagnostic)**.** Give a rigorous worked analysis of Nonnegative functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For measurable $f \ge 0$, define the integral as the supremum of simple-function integrals over $0 \le \phi \le f$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 3.6 (Positive and negative parts diagnostic)**.** Give a rigorous worked analysis of Positive and negative parts in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Write $f = f^+ - f^-$ with $f^+, f^- \ge 0$ and $|f| = f^+ + f^-$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 3.7 (Integrable functions diagnostic)**.** Give a rigorous worked analysis of Integrable functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A measurable real-valued function is integrable when $\int |f| < \infty$, which rules out the indeterminate expression $\infty - \infty$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

3.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 3.8 (Simple-function integral diagnostic)**.** Give a rigorous worked analysis of Simple-function integral in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For a nonnegative simple function $\phi = \sum a_k 1_{E_k}$ on disjoint measurable level sets, define $\int \phi = \sum a_k \mu(E_k)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 3.9 (Nonnegative functions diagnostic)**.** Give a rigorous worked analysis of Nonnegative functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For measurable $f \geq 0$, define the integral as the supremum of simple-function integrals over $0 \leq \phi \leq f$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 3.10 (Positive and negative parts diagnostic)**.** Give a rigorous worked analysis of Positive and negative parts in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Write $f = f^+ - f^-$ with $f^+, f^- \geq 0$ and $|f| = f^+ + f^-$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 3.11 (Integrable functions diagnostic)**.** Give a rigorous worked analysis of Integrable functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A measurable real-valued function is integrable when $\int |f| < \infty$, which rules out the indeterminate expression $\infty - \infty$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 3.12 (Monotonicity diagnostic)**.** Give a rigorous worked analysis of Monotonicity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. If $0 \leq f \leq g$, then $\int f \leq \int g$ because every simple lower approximation to f is also one for g. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 3.13 (Linearity diagnostic)**.** Give a rigorous worked analysis of Linearity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Integrable functions form a vector space and the integral is linear. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 3.14 (Null-set invariance diagnostic)**.** Give a rigorous worked analysis of Null-set invariance in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. If $f = g$ almost everywhere and one is integrable, then both represent the same integral. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 3.15 (Integrals over sets diagnostic)**.** Give a rigorous worked analysis of Integrals over sets in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Define $\int_E f = \int f 1_E$, making restriction to measurable subsets part of the same formalism. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 3.16 (Absolute integral inequality proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For integrable f, $|\int f| \leq \int |f|$.

Solution. Use $-|f| \leq f \leq |f|$ and monotonicity of the integral. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 3.17 (Zero integral criterion proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f \geq 0$ and $\int f = 0$, then $f = 0$ almost everywhere.

Solution. For $E_n = \{f \geq 1/n\}$, the inequality $\int f \geq (1/n)\mu(E_n)$ forces every E_n null; their union is $\{f > 0\}$. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 3.18 (Linearity proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If f, g are integrable and $a, b \in \mathbb{R}$, then $\int(af + bg) = a\int f + b\int g$.

Solution. Prove first for nonnegative simple functions on a common refinement, pass to nonnegative measurable functions by monotone approximation, then use positive/negative parts. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 3.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. The Lebesgue integral is constructed by integrating simple functions and taking monotone suprema. It is insensitive to null-set changes, linear on integrable functions, and controlled by the integral of the absolute value. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

3.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 3.20. State and apply the central principle behind Simple-function integral. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. For a nonnegative simple function $\phi = \sum a_k 1_{E_k}$ on disjoint measurable level sets, define $\int \phi = \sum a_k \mu(E_k)$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 3.21. State and apply the central principle behind Nonnegative functions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. For measurable $f \geq 0$, define the integral as the supremum of simple-function integrals over $0 \leq \phi \leq f$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 3.22. State and apply the central principle behind Positive and negative parts. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Write $f = f^+ - f^-$ with $f^+, f^- \geq 0$ and $|f| = f^+ + f^-$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 3.23. State and apply the central principle behind Integrable functions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. A measurable real-valued function is integrable when $\int |f| < \infty$, which rules out the indeterminate expression $\infty - \infty$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 3.24. State and apply the central principle behind Monotonicity. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. If $0 \leq f \leq g$, then $\int f \leq \int g$ because every simple lower approximation to f is also one for g. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 3.25. State and apply the central principle behind Linearity. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Integrable functions form a vector space and the integral is linear. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 3.26. State and apply the central principle behind Null-set invariance. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. If $f = g$ almost everywhere and one is integrable, then both represent the same integral. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 3.27. State and apply the central principle behind Integrals over sets. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Define $\int_E f = \int f 1_E$, making restriction to measurable subsets part of the same formalism. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 4

Convergence Theorems

Monotone convergence, Fatou's lemma, and dominated convergence govern the passage of limits through integrals. Together they separate the roles of positivity, one-sided control, and integrable domination.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

4.1 Monotone convergence

Increasing nonnegative sequences allow direct interchange of pointwise limit and integral.

4.2 Fatou's lemma

The integral of a nonnegative liminf cannot exceed the liminf of the integrals.

4.3 Dominated convergence

Almost-everywhere convergence under a single integrable absolute bound permits limit-integral interchange for signed functions.

4.4 Almost-everywhere convergence

Null exceptional sets do not affect the integral but still require measurable representatives.

4.5 Convergence in measure

Error sets $\{|f_n - f| > \varepsilon\}$ shrinking in measure define a weaker convergence mode.

4.6 L^1 convergence

Norm convergence controls the integral of absolute error and implies convergence in measure.

4.7 Series applications

MCT handles nonnegative series through increasing partial sums; DCT handles dominated signed series.

4.8 Counterexamples

Moving spikes show that pointwise convergence alone does not control integrals.

4.9 Core structural results

Theorem 4.1 (Monotone convergence theorem)**.** *If $0 \le f_n \uparrow f$, then $\int f_n \uparrow \int f$.*

Proof. The integrals increase and are bounded above by $\int f$; every simple function below f is eventually captured up to an arbitrarily small factor on most of its level sets, forcing equality of the limit. □

Theorem 4.2 (Fatou's lemma)**.** *For nonnegative measurable f_n, $\int \liminf f_n \le \liminf \int f_n$.*

Proof. Set $g_N = \inf_{n \ge N} f_n$. Then $g_N \uparrow \liminf f_n$ and $\int g_N \le \inf_{n \ge N} \int f_n$; apply MCT and let $N \to \infty$. □

Theorem 4.3 (Dominated convergence theorem)**.** *If $f_n \to f$ a.e. and $|f_n| \le g \in L^1$, then $\int f_n \to \int f$.*

Proof. Apply Fatou to the nonnegative sequences $g \pm f_n$ to obtain matching upper and lower bounds; also $|f| \le g$ a.e., so f is integrable. □

4.10 Worked examples

Example 4.4 (Monotone convergence diagnostic)**.** Give a rigorous worked analysis of Monotone convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Increasing nonnegative sequences allow direct interchange of pointwise limit and integral. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 4.5 (Fatou's lemma diagnostic)**.** Give a rigorous worked analysis of Fatou's lemma in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The integral of a nonnegative liminf cannot exceed the liminf of the integrals. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 4.6 (Dominated convergence diagnostic)**.** Give a rigorous worked analysis of Dominated convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Almost-everywhere convergence under a single integrable absolute bound permits limit-integral interchange for signed functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 4.7 (Almost-everywhere convergence diagnostic)**.** Give a rigorous worked analysis of Almost-everywhere convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Null exceptional sets do not affect the integral but still require measurable representatives. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

4.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 4.8 (Monotone convergence diagnostic)**.** Give a rigorous worked analysis of Monotone convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Increasing nonnegative sequences allow direct interchange of pointwise limit and integral. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 4.9 (Fatou's lemma diagnostic)**.** Give a rigorous worked analysis of Fatou's lemma in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The integral of a nonnegative liminf cannot exceed the liminf of the integrals. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 4.10 (Dominated convergence diagnostic)**.** Give a rigorous worked analysis of Dominated convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Almost-everywhere convergence under a single integrable absolute bound permits limit-integral interchange for signed functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 4.11 (Almost-everywhere convergence diagnostic)**.** Give a rigorous worked analysis of Almost-everywhere convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Null exceptional sets do not affect the integral but still require measurable representatives. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 4.12 (Convergence in measure diagnostic)**.** Give a rigorous worked analysis of Convergence in measure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Error sets $\{|f_n - f| > \varepsilon\}$ shrinking in measure define a weaker convergence mode. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 4.13 (L^1 convergence diagnostic)**.** Give a rigorous worked analysis of L^1 convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Norm convergence controls the integral of absolute error and implies convergence in measure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 4.14 (Series applications diagnostic)**.** Give a rigorous worked analysis of Series applications in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. MCT handles nonnegative series through increasing partial sums; DCT handles dominated signed series. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 4.15 (Counterexamples diagnostic)**.** Give a rigorous worked analysis of Counterexamples in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Moving spikes show that pointwise convergence alone does not control integrals. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 4.16 (Monotone convergence theorem proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $0 \leq f_n \uparrow f$, then $\int f_n \uparrow \int f$.

Solution. The integrals increase and are bounded above by $\int f$; every simple function below f is eventually captured up to an arbitrarily small factor on most of its level sets, forcing equality of the limit. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 4.17 (Fatou's lemma proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For nonnegative measurable f_n, $\int \liminf f_n \leq \liminf \int f_n$.

Solution. Set $g_N = \inf_{n \geq N} f_n$. Then $g_N \uparrow \liminf f_n$ and $\int g_N \leq \inf_{n \geq N} \int f_n$; apply MCT and let $N \to \infty$. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 4.18 (Dominated convergence theorem proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f_n \to f$ a.e. and $|f_n| \leq g \in L^1$, then $\int f_n \to \int f$.

Solution. Apply Fatou to the nonnegative sequences $g \pm f_n$ to obtain matching upper and lower bounds; also $|f| \leq g$ a.e., so f is integrable. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 4.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Monotone convergence, Fatou's lemma, and dominated convergence govern the passage of limits through integrals. Together they separate the roles of positivity, one-sided control, and integrable domination. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

4.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 4.20. State and apply the central principle behind Monotone convergence. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Increasing nonnegative sequences allow direct interchange of pointwise limit and integral. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 4.21. State and apply the central principle behind Fatou's lemma. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The integral of a nonnegative liminf cannot exceed the liminf of the integrals. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 4.22. State and apply the central principle behind Dominated convergence. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Almost-everywhere convergence under a single integrable absolute bound permits limit-integral interchange for signed functions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 4.23. State and apply the central principle behind Almost-everywhere convergence. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Null exceptional sets do not affect the integral but still require measurable representatives. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 4.24. State and apply the central principle behind Convergence in measure. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Error sets $\{|f_n - f| > \varepsilon\}$ shrinking in measure define a weaker convergence mode. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 4.25. State and apply the central principle behind L^1 convergence. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Norm convergence controls the integral of absolute error and implies convergence in measure. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 4.26. State and apply the central principle behind Series applications. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. MCT handles nonnegative series through increasing partial sums; DCT handles dominated signed series. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 4.27. State and apply the central principle behind Counterexamples. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Moving spikes show that pointwise convergence alone does not control integrals. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 5

Product Measures and Fubini Theory

Product measures combine measurable coordinate spaces, while Tonelli and Fubini convert multidimensional integration into iterated integration. Absolute integrability is the key condition that permits changing order safely.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

5.1 Product sigma-algebra

The product sigma-algebra is generated by measurable rectangles $A \times B$.

5.2 Product measures

On sigma-finite spaces, the product measure is characterized by $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$ on rectangles.

5.3 Sections of sets

For measurable $E \subset X \times Y$, the section $E_x = \{y : (x, y) \in E\}$ is measurable for almost every relevant parameter under the standard product theory.

5.4 Sections of functions

Fixing one variable in a measurable function produces measurable sections, allowing iterated integration.

5.5 Tonelli theorem

Nonnegative measurable functions may be integrated in either order, with the common value possibly infinite.

5.6 Fubini theorem

Absolutely integrable functions have integrable sections almost everywhere and finite iterated integrals equal to the product integral.

5.7 Changing order

Absolute integrability prevents order-dependent cancellation.

5.8 Applications

Areas, convolution kernels, expectations, and integral operators all rely on product integration.

5.9 Core structural results

Theorem 5.1 (Tonelli theorem)**.** *For nonnegative measurable f on a sigma-finite product space, the product integral equals both iterated integrals, possibly with value ∞.*

Proof. Prove first for rectangle indicators, then nonnegative simple functions, and finally pass to arbitrary nonnegative measurable functions by monotone convergence. ☐

Theorem 5.2 (Fubini theorem)**.** *If $f \in L^1(\mu \times \nu)$, then almost all sections are integrable and the iterated integrals equal $\int f$.*

Proof. Apply Tonelli to $|f|$ to obtain finite section integrals almost everywhere, then decompose f into positive and negative parts. ☐

Theorem 5.3 (Rectangle rule)**.** *For measurable rectangles, $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$.*

Proof. This defines the canonical premeasure on rectangles; sigma-finiteness gives the standard extension and uniqueness on the generated product sigma-algebra. ☐

5.10 Worked examples

Example 5.4 (Product sigma-algebra diagnostic)**.** Give a rigorous worked analysis of Product sigma-algebra in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The product sigma-algebra is generated by measurable rectangles $A \times B$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 5.5 (Product measures diagnostic)**.** Give a rigorous worked analysis of Product measures in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. On sigma-finite spaces, the product measure is characterized by $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$ on rectangles. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 5.6 (Sections of sets diagnostic)**.** Give a rigorous worked analysis of Sections of sets in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For measurable $E \subset X \times Y$, the section $E_x = \{y : (x, y) \in E\}$ is measurable for almost every relevant parameter under the standard product theory. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 5.7 (Sections of functions diagnostic)**.** Give a rigorous worked analysis of Sections of functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Fixing one variable in a measurable function produces measurable sections, allowing iterated integration. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

5.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 5.8 (Product sigma-algebra diagnostic)**.** Give a rigorous worked analysis of Product sigma-algebra in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The product sigma-algebra is generated by measurable rectangles $A \times B$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 5.9 (Product measures diagnostic)**.** Give a rigorous worked analysis of Product measures in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. On sigma-finite spaces, the product measure is characterized by $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$ on rectangles. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 5.10 (Sections of sets diagnostic)**.** Give a rigorous worked analysis of Sections of sets in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For measurable $E \subset X \times Y$, the section $E_x = \{y : (x, y) \in E\}$ is measurable for almost every relevant parameter under the standard product theory. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 5.11 (Sections of functions diagnostic)**.** Give a rigorous worked analysis of Sections of functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Fixing one variable in a measurable function produces measurable sections, allowing iterated integration. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 5.12 (Tonelli theorem diagnostic)**.** Give a rigorous worked analysis of Tonelli theorem in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Nonnegative measurable functions may be integrated in either order, with the common value possibly infinite. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 5.13 (Fubini theorem diagnostic)**.** Give a rigorous worked analysis of Fubini theorem in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Absolutely integrable functions have integrable sections almost everywhere and finite iterated integrals equal to the product integral. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 5.14 (Changing order diagnostic)**.** Give a rigorous worked analysis of Changing order in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Absolute integrability prevents order-dependent cancellation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 5.15 (Applications diagnostic)**.** Give a rigorous worked analysis of Applications in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Areas, convolution kernels, expectations, and integral operators all rely on product integration. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 5.16 (Tonelli theorem proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For nonnegative measurable f on a sigma-finite product space, the product integral equals both iterated integrals, possibly with value ∞.

Solution. Prove first for rectangle indicators, then nonnegative simple functions, and finally pass to arbitrary nonnegative measurable functions by monotone convergence. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 5.17 (Fubini theorem proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in L^1(\mu \times \nu)$, then almost all sections are integrable and the iterated integrals equal $\int f$.

Solution. Apply Tonelli to $|f|$ to obtain finite section integrals almost everywhere, then decompose f into positive and negative parts. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 5.18 (Rectangle rule proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For measurable rectangles, $(\mu\times\nu)(A\times B) = \mu(A)\nu(B)$.

Solution. This defines the canonical premeasure on rectangles; sigma-finiteness gives the standard extension and uniqueness on the generated product sigma-algebra. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 5.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Product measures combine measurable coordinate spaces, while Tonelli and Fubini convert multidimensional integration into iterated integration. Absolute integrability is the key condition that permits changing order safely. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. ☐

5.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 5.20. State and apply the central principle behind Product sigma-algebra. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. The product sigma-algebra is generated by measurable rectangles $A \times B$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 5.21. State and apply the central principle behind Product measures. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. On sigma-finite spaces, the product measure is characterized by $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$ on rectangles. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 5.22. State and apply the central principle behind Sections of sets. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. For measurable $E \subset X \times Y$, the section $E_x = \{y : (x, y) \in E\}$ is measurable for almost every relevant parameter under the standard product theory. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 5.23. State and apply the central principle behind Sections of functions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Fixing one variable in a measurable function produces measurable sections, allowing iterated integration. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 5.24. State and apply the central principle behind Tonelli theorem. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Nonnegative measurable functions may be integrated in either order, with the common value possibly infinite. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 5.25. State and apply the central principle behind Fubini theorem. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Absolutely integrable functions have integrable sections almost everywhere and finite iterated integrals equal to the product integral. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 5.26. State and apply the central principle behind Changing order. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Absolute integrability prevents order-dependent cancellation. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 5.27. State and apply the central principle behind Applications. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Areas, convolution kernels, expectations, and integral operators all rely on product integration. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 6

Lp Spaces

L^p spaces turn integral size into geometry. After identifying functions equal almost everywhere, the L^p quantities become genuine norms for $p \geq 1$, and the resulting spaces are complete.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

6.1 Almost-everywhere classes

Two measurable functions equal almost everywhere represent the same L^p element.

6.2 L^p norms

For $1 \leq p < \infty$, use $\|f\|_p = (\int |f|^p)^{1/p}$; for $p = \infty$, use essential supremum.

6.3 Essential supremum

The L^∞ norm ignores exceptional values on null sets and records the least almost-everywhere uniform bound.

6.4 Vector-space structure

Hölder and Minkowski control sums and scalar multiples, making L^p a normed vector space.

6.5 Completeness

Every L^p Cauchy sequence converges in L^p for $1 \leq p \leq \infty$.

6.6 Finite-measure inclusions

On finite-measure spaces, $L^q \subset L^p$ when $q > p$, with an explicit measure factor.

6.7 Convergence in measure

For finite p, L^p convergence implies convergence in measure by Chebyshev's inequality.

6.8 Approximation

Truncation, finite-measure cutoffs, and simple functions provide dense approximations in L^p for $p < \infty$.

6.9 Core structural results

Theorem 6.1 (Norm-zero criterion). *For $1 \le p \le \infty$, $\|f\|_p = 0$ iff $f = 0$ almost everywhere.*

Proof. For finite p, $\int |f|^p = 0$ forces $|f|^p = 0$ a.e.; for $p = \infty$, essential supremum zero means $|f| = 0$ a.e. □

Theorem 6.2 (Finite-measure inclusion). *If $\mu(X) < \infty$ and $1 \le p < q \le \infty$, then $\|f\|_p \le \mu(X)^{1/p-1/q}\|f\|_q$.*

Proof. For finite q, apply Hölder to $|f|^p \cdot 1$ with exponent q/p and its conjugate; the $q = \infty$ case follows directly. □

Theorem 6.3 (Completeness of L^p). *For $1 \le p \le \infty$, L^p is complete.*

Proof. Extract a subsequence with summable norm increments, use the resulting pointwise absolutely summable majorant to construct an a.e. limit, prove subsequence norm convergence, and recover the full Cauchy sequence. □

6.10 Worked examples

Example 6.4 (Almost-everywhere classes diagnostic). Give a rigorous worked analysis of Almost-everywhere classes in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Two measurable functions equal almost everywhere represent the same L^p element. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 6.5 (L^p norms diagnostic). Give a rigorous worked analysis of L^p norms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For $1 \le p < \infty$, use $\|f\|_p = (\int |f|^p)^{1/p}$; for $p = \infty$, use essential supremum. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 6.6 (Essential supremum diagnostic). Give a rigorous worked analysis of Essential supremum in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The L^∞ norm ignores exceptional values on null sets and records the least almost-everywhere uniform bound. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 6.7 (Vector-space structure diagnostic)**.** Give a rigorous worked analysis of Vector-space structure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Hölder and Minkowski control sums and scalar multiples, making L^p a normed vector space. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

6.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 6.8 (Almost-everywhere classes diagnostic)**.** Give a rigorous worked analysis of Almost-everywhere classes in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Two measurable functions equal almost everywhere represent the same L^p element. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 6.9 (L^p norms diagnostic)**.** Give a rigorous worked analysis of L^p norms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For $1 \le p < \infty$, use $\|f\|_p = (\int |f|^p)^{1/p}$; for $p = \infty$, use essential supremum. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 6.10 (Essential supremum diagnostic)**.** Give a rigorous worked analysis of Essential supremum in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The L^∞ norm ignores exceptional values on null sets and records the least almost-everywhere uniform bound. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 6.11 (Vector-space structure diagnostic)**.** Give a rigorous worked analysis of Vector-space structure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Hölder and Minkowski control sums and scalar multiples, making L^p a normed vector space. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 6.12 (Completeness diagnostic)**.** Give a rigorous worked analysis of Completeness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Every L^p Cauchy sequence converges in L^p for $1 \le p \le \infty$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 6.13 (Finite-measure inclusions diagnostic)**.** Give a rigorous worked analysis of Finite-measure inclusions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. On finite-measure spaces, $L^q \subset L^p$ when $q > p$, with an explicit measure factor. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 6.14 (Convergence in measure diagnostic)**.** Give a rigorous worked analysis of Convergence in measure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For finite p, L^p convergence implies convergence in measure by Chebyshev's inequality. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 6.15 (Approximation diagnostic)**.** Give a rigorous worked analysis of Approximation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Truncation, finite-measure cutoffs, and simple functions provide dense approximations in L^p for $p < \infty$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 6.16 (Norm-zero criterion proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For $1 \leq p \leq \infty$, $\|f\|_p = 0$ iff $f = 0$ almost everywhere.

Solution. For finite p, $\int |f|^p = 0$ forces $|f|^p = 0$ a.e.; for $p = \infty$, essential supremum zero means $|f| = 0$ a.e. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 6.17 (Finite-measure inclusion proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $\mu(X) < \infty$ and $1 \leq p < q \leq \infty$, then $\|f\|_p \leq \mu(X)^{1/p-1/q}\|f\|_q$.

Solution. For finite q, apply Hölder to $|f|^p \cdot 1$ with exponent q/p and its conjugate; the $q = \infty$ case follows directly. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 6.18 (Completeness of L^p proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For $1 \leq p \leq \infty$, L^p is complete.

Solution. Extract a subsequence with summable norm increments, use the resulting pointwise absolutely summable majorant to construct an a.e. limit, prove subsequence norm convergence, and recover the full Cauchy sequence. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 6.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. L^p spaces turn integral size into geometry. After identifying functions equal almost everywhere, the L^p quantities become genuine norms for $p \geq 1$, and the resulting spaces are complete. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

6.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 6.20. State and apply the central principle behind Almost-everywhere classes. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Two measurable functions equal almost everywhere represent the same L^p element. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 6.21. State and apply the central principle behind L^p norms. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. For $1 \le p < \infty$, use $\|f\|_p = (\int |f|^p)^{1/p}$; for $p = \infty$, use essential supremum. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 6.22. State and apply the central principle behind Essential supremum. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. The L^∞ norm ignores exceptional values on null sets and records the least almost-everywhere uniform bound. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 6.23. State and apply the central principle behind Vector-space structure. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Hölder and Minkowski control sums and scalar multiples, making L^p a normed vector space. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 6.24. State and apply the central principle behind Completeness. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Every L^p Cauchy sequence converges in L^p for $1 \le p \le \infty$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 6.25. State and apply the central principle behind Finite-measure inclusions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. On finite-measure spaces, $L^q \subset L^p$ when $q > p$, with an explicit measure factor. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 6.26. State and apply the central principle behind Convergence in measure. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. For finite p, L^p convergence implies convergence in measure by Chebyshev's inequality. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 6.27. State and apply the central principle behind Approximation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Truncation, finite-measure cutoffs, and simple functions provide dense approximations in L^p for $p < \infty$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 7

Hölder, Minkowski and Interpolation

Hölder and Minkowski are the fundamental inequalities of L^p geometry. Interpolation then explains how information at two exponents controls intermediate exponents.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

7.1 Conjugate exponents

For $1 < p < \infty$, the conjugate exponent q satisfies $1/p + 1/q = 1$.

7.2 Young inequality

The scalar bound $ab \leq a^p/p + b^q/q$ encodes the convex duality behind Hölder.

7.3 Hölder inequality

Products of L^p and L^q functions lie in L^1 with sharp norm control.

7.4 Minkowski inequality

The L^p norm satisfies the triangle inequality.

7.5 Equality cases

Sharpness is governed by proportionality of the powers $|f|^p$ and $|g|^q$.

7.6 Discrete versions

Counting measure recovers the familiar finite-sum Hölder and Minkowski inequalities.

7.7 Interpolation of norms

Intermediate L^r norms are logarithmically convex between endpoint norms.

7.8 Operator interpolation preview

Linear operators bounded at endpoint exponents often inherit intermediate bounds under an interpolation theorem.

7.9 Core structural results

Theorem 7.1 (Hölder inequality). *If $1 < p, q < \infty$ are conjugate, then $\int |fg| \le \|f\|_p \|g\|_q$.*

Proof. Normalize the two norms to one, apply Young's inequality pointwise to the normalized product, integrate, and rescale. ☐

Theorem 7.2 (Minkowski inequality). *For $1 \le p < \infty$, $\|f + g\|_p \le \|f\|_p + \|g\|_p$.*

Proof. For $p > 1$, estimate $\int |f + g|^p$ by pairing $|f + g|^{p-1}$ with $|f| + |g|$ and apply Hölder; the case $p = 1$ is immediate. ☐

Theorem 7.3 (Log-convex interpolation). *If $1/r = \theta/p + (1-\theta)/q$, then $\|f\|_r \le \|f\|_p^{\theta} \|f\|_q^{1-\theta}$.*

Proof. Factor $|f|^r$ into suitable powers of $|f|^p$ and $|f|^q$ and apply Hölder with reciprocal weights. ☐

7.10 Worked examples

Example 7.4 (Conjugate exponents diagnostic). Give a rigorous worked analysis of Conjugate exponents in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For $1 < p < \infty$, the conjugate exponent q satisfies $1/p + 1/q = 1$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 7.5 (Young inequality diagnostic). Give a rigorous worked analysis of Young inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The scalar bound $ab \le a^p/p + b^q/q$ encodes the convex duality behind Hölder. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 7.6 (Hölder inequality diagnostic). Give a rigorous worked analysis of Hölder inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Products of L^p and L^q functions lie in L^1 with sharp norm control. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 7.7 (Minkowski inequality diagnostic)**.** Give a rigorous worked analysis of Minkowski inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The L^p norm satisfies the triangle inequality. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

7.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 7.8 (Conjugate exponents diagnostic)**.** Give a rigorous worked analysis of Conjugate exponents in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For $1 < p < \infty$, the conjugate exponent q satisfies $1/p+1/q = 1$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 7.9 (Young inequality diagnostic)**.** Give a rigorous worked analysis of Young inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The scalar bound $ab \leq a^p/p + b^q/q$ encodes the convex duality behind Hölder. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 7.10 (Hölder inequality diagnostic)**.** Give a rigorous worked analysis of Hölder inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Products of L^p and L^q functions lie in L^1 with sharp norm control. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 7.11 (Minkowski inequality diagnostic)**.** Give a rigorous worked analysis of Minkowski inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The L^p norm satisfies the triangle inequality. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 7.12 (Equality cases diagnostic)**.** Give a rigorous worked analysis of Equality cases in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Sharpness is governed by proportionality of the powers $|f|^p$ and $|g|^q$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 7.13 (Discrete versions diagnostic)**.** Give a rigorous worked analysis of Discrete versions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Counting measure recovers the familiar finite-sum Hölder and Minkowski inequalities. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 7.14 (Interpolation of norms diagnostic)**.** Give a rigorous worked analysis of Interpolation of norms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Intermediate L^r norms are logarithmically convex between endpoint norms. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 7.15 (Operator interpolation preview diagnostic)**.** Give a rigorous worked analysis of Operator interpolation preview in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Linear operators bounded at endpoint exponents often inherit intermediate bounds under an interpolation theorem. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 7.16 (Hölder inequality proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $1 < p, q < \infty$ are conjugate, then $\int |fg| \leq \|f\|_p \|g\|_q$.

Solution. Normalize the two norms to one, apply Young's inequality pointwise to the normalized product, integrate, and rescale. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 7.17 (Minkowski inequality proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For $1 \leq p < \infty$, $\|f+g\|_p \leq \|f\|_p + \|g\|_p$.

Solution. For $p > 1$, estimate $\int |f+g|^p$ by pairing $|f+g|^{p-1}$ with $|f| + |g|$ and apply Hölder; the case $p = 1$ is immediate. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 7.18 (Log-convex interpolation proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $1/r = \theta/p + (1-\theta)/q$, then $\|f\|_r \leq \|f\|_p^{\theta} \|f\|_q^{1-\theta}$.

Solution. Factor $|f|^r$ into suitable powers of $|f|^p$ and $|f|^q$ and apply Hölder with reciprocal weights. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 7.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Hölder and Minkowski are the fundamental inequalities of L^p geometry. Interpolation then explains how information at two exponents controls intermediate exponents. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

7.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 7.20. State and apply the central principle behind Conjugate exponents. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. For $1 < p < \infty$, the conjugate exponent q satisfies $1/p + 1/q = 1$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 7.21. State and apply the central principle behind Young inequality. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The scalar bound $ab \leq a^p/p + b^q/q$ encodes the convex duality behind Hölder. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 7.22. State and apply the central principle behind Hölder inequality. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Products of L^p and L^q functions lie in L^1 with sharp norm control. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 7.23. State and apply the central principle behind Minkowski inequality. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The L^p norm satisfies the triangle inequality. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 7.24. State and apply the central principle behind Equality cases. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Sharpness is governed by proportionality of the powers $|f|^p$ and $|g|^q$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 7.25. State and apply the central principle behind Discrete versions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Counting measure recovers the familiar finite-sum Hölder and Minkowski inequalities. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 7.26. State and apply the central principle behind Interpolation of norms. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Intermediate L^r norms are logarithmically convex between endpoint norms. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 7.27. State and apply the central principle behind Operator interpolation preview. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Linear operators bounded at endpoint exponents often inherit intermediate bounds under an interpolation theorem. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 8

Egorov, Vitali and Weak-Lp Ideas

Egorov, Vitali, and weak-L^p theory compare convergence modes and exceptional-set control. They clarify how almost-everywhere convergence can become nearly uniform and how tail bounds encode borderline integrability.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

8.1 Convergence in measure

A sequence converges in measure when $\mu(|f_n - f| > \varepsilon) \to 0$ for every $\varepsilon > 0$.

8.2 Egorov theorem

On finite-measure spaces, almost-everywhere convergence becomes uniform after removing a set of arbitrarily small measure.

8.3 Uniform integrability

A family is uniformly integrable when integral mass cannot concentrate on sets of very small measure.

8.4 Vitali convergence

Convergence in measure plus uniform integrability upgrades to L^1 convergence under the standard finite-measure hypotheses.

8.5 Weak-L^p

Weak-L^p controls tails through $\sup_{t>0} t^p \mu(|f| > t)$ rather than directly integrating $|f|^p$.

8.6 Chebyshev inequality

Strong L^p bounds imply weak-L^p tail estimates.

8.7 Subsequence extraction

Convergence in measure admits almost-everywhere convergent subsequences under standard assumptions.

8.8 Borderline behavior

Power singularities often lie in weak-L^p at the exact exponent where strong L^p fails.

8.9 Core structural results

Theorem 8.1 (Egorov theorem)**.** *If $\mu(X) < \infty$ and $f_n \to f$ a.e., then for every $\varepsilon > 0$ there is measurable E with $\mu(E) < \varepsilon$ such that convergence is uniform on $X \setminus E$.*

Proof. For each accuracy level, form decreasing tail-exception sets. Their intersection is null; continuity from above on the finite-measure space yields a late stage with small measure, and a diagonal choice handles all accuracy levels. ☐

Theorem 8.2 (Chebyshev inequality)**.** *If $f \in L^p$ and $t > 0$, then $\mu(|f| > t) \leq t^{-p}\|f\|_p^p$.*

Proof. On the superlevel set, $|f|^p \geq t^p$; integrate and rearrange. ☐

Theorem 8.3 (Vitali convergence principle)**.** *On a finite-measure space, convergence in measure plus uniform integrability implies L^1 convergence.*

Proof. Split the error integral into the region where the error is small and its exceptional set; convergence in measure makes the exceptional set small and uniform integrability controls the mass there. ☐

8.10 Worked examples

Example 8.4 (Convergence in measure diagnostic)**.** Give a rigorous worked analysis of Convergence in measure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A sequence converges in measure when $\mu(|f_n - f| > \varepsilon) \to 0$ for every $\varepsilon > 0$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 8.5 (Egorov theorem diagnostic)**.** Give a rigorous worked analysis of Egorov theorem in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. On finite-measure spaces, almost-everywhere convergence becomes uniform after removing a set of arbitrarily small measure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 8.6 (Uniform integrability diagnostic)**.** Give a rigorous worked analysis of Uniform integrability in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A family is uniformly integrable when integral mass cannot concentrate on sets of very small measure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 8.7 (Vitali convergence diagnostic)**.** Give a rigorous worked analysis of Vitali convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Convergence in measure plus uniform integrability upgrades to L^1 convergence under the standard finite-measure hypotheses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

8.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 8.8 (Convergence in measure diagnostic)**.** Give a rigorous worked analysis of Convergence in measure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A sequence converges in measure when $\mu(|f_n - f| > \varepsilon) \to 0$ for every $\varepsilon > 0$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 8.9 (Egorov theorem diagnostic)**.** Give a rigorous worked analysis of Egorov theorem in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. On finite-measure spaces, almost-everywhere convergence becomes uniform after removing a set of arbitrarily small measure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 8.10 (Uniform integrability diagnostic)**.** Give a rigorous worked analysis of Uniform integrability in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A family is uniformly integrable when integral mass cannot concentrate on sets of very small measure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 8.11 (Vitali convergence diagnostic)**.** Give a rigorous worked analysis of Vitali convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Convergence in measure plus uniform integrability upgrades to L^1 convergence under the standard finite-measure hypotheses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 8.12 (Weak-L^p diagnostic)**.** Give a rigorous worked analysis of Weak-L^p in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Weak-L^p controls tails through $\sup_{t>0} t^p \mu(|f| > t)$ rather than directly integrating $|f|^p$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 8.13 (Chebyshev inequality diagnostic)**.** Give a rigorous worked analysis of Chebyshev inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Strong L^p bounds imply weak-L^p tail estimates. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 8.14 (Subsequence extraction diagnostic)**.** Give a rigorous worked analysis of Subsequence extraction in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Convergence in measure admits almost-everywhere convergent subsequences under standard assumptions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 8.15 (Borderline behavior diagnostic)**.** Give a rigorous worked analysis of Borderline behavior in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Power singularities often lie in weak-L^p at the exact exponent where strong L^p fails. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 8.16 (Egorov theorem proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $\mu(X) < \infty$ and $f_n \to f$ a.e., then for every $\varepsilon > 0$ there is measurable E with $\mu(E) < \varepsilon$ such that convergence is uniform on $X \setminus E$.

Solution. For each accuracy level, form decreasing tail-exception sets. Their intersection is null; continuity from above on the finite-measure space yields a late stage with small measure, and a diagonal choice handles all accuracy levels. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 8.17 (Chebyshev inequality proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in L^p$ and $t > 0$, then $\mu(|f| > t) \le t^{-p}\|f\|_p^p$.

Solution. On the superlevel set, $|f|^p \ge t^p$; integrate and rearrange. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 8.18 (Vitali convergence principle proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: On a finite-measure space, convergence in measure plus uniform integrability implies L^1 convergence.

Solution. Split the error integral into the region where the error is small and its exceptional set; convergence in measure makes the exceptional set small and uniform integrability controls the mass there. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 8.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Egorov, Vitali, and weak-L^p theory compare convergence modes and exceptional-set control. They clarify how almost-everywhere convergence can become nearly uniform and how tail bounds encode borderline integrability. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

8.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 8.20. State and apply the central principle behind Convergence in measure. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A sequence converges in measure when $\mu(|f_n - f| > \varepsilon) \to 0$ for every $\varepsilon > 0$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 8.21. State and apply the central principle behind Egorov theorem. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. On finite-measure spaces, almost-everywhere convergence becomes uniform after removing a set of arbitrarily small measure. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 8.22. State and apply the central principle behind Uniform integrability. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A family is uniformly integrable when integral mass cannot concentrate on sets of very small measure. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 8.23. State and apply the central principle behind Vitali convergence. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Convergence in measure plus uniform integrability upgrades to L^1 convergence under the standard finite-measure hypotheses. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 8.24. State and apply the central principle behind Weak-L^p. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Weak-L^p controls tails through $\sup_{t>0} t^p\mu(|f| > t)$ rather than directly integrating $|f|^p$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 8.25. State and apply the central principle behind Chebyshev inequality. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Strong L^p bounds imply weak-L^p tail estimates. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 8.26. State and apply the central principle behind Subsequence extraction. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Convergence in measure admits almost-everywhere convergent subsequences under standard assumptions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 8.27. State and apply the central principle behind Borderline behavior. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Power singularities often lie in weak-L^p at the exact exponent where strong L^p fails. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Part II

Fourier Analysis

Chapter 9

Fourier Series

Fourier series decompose periodic functions into orthogonal trigonometric modes. Coefficients are projections, partial sums are finite-dimensional approximations, and convergence depends on both function regularity and the geometry of the trigonometric system.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

9.1 Periodic functions

A 2π-periodic function can be viewed as a function on the circle, making trigonometric modes natural coordinates.

9.2 Orthogonality

Distinct sine and cosine modes are orthogonal on a full period.

9.3 Fourier coefficients

Coefficients are obtained by integrating against the corresponding orthogonal modes.

9.4 Partial sums

The Nth Fourier partial sum is the orthogonal projection onto trigonometric polynomials of degree at most N.

9.5 Bessel inequality

Finite coefficient energy cannot exceed the L^2 energy of the function.

9.6 Dirichlet kernel

Partial sums can be written as convolution with the Dirichlet kernel, making convergence an approximate-identity problem with oscillatory kernels.

9.7 Pointwise convergence

Piecewise smooth functions converge to their value at continuity points and to midpoint values at jumps under classical Dirichlet hypotheses.

9.8 Gibbs phenomenon

Near jumps, overshoot persists in relative height while its spatial width shrinks.

9.9 Core structural results

Theorem 9.1 (Orthogonality relations)**.** *Distinct trigonometric modes are orthogonal in* $L^2[-\pi, \pi]$.

Proof. Use product-to-sum identities; every nonconstant integer-frequency sine or cosine integrates to zero over a full period. ☐

Theorem 9.2 (Bessel inequality)**.** *The square-sum of Fourier coefficients over any finite set of orthonormal modes is bounded by* $\|f\|_2^2$.

Proof. Subtract the finite orthogonal projection from f and expand the nonnegative squared norm using orthogonality. ☐

Theorem 9.3 (Dirichlet convergence principle)**.** *Under standard piecewise smooth hypotheses, Fourier partial sums converge at* x *to* $(f(x^-) + f(x^+))/2$.

Proof. Express the partial sum by the Dirichlet kernel, split the integral into a small neighborhood and its complement, use local one-sided regularity near zero and oscillatory cancellation away from zero. ☐

9.10 Worked examples

Example 9.4 (Periodic functions diagnostic)**.** Give a rigorous worked analysis of Periodic functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A 2π-periodic function can be viewed as a function on the circle, making trigonometric modes natural coordinates. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 9.5 (Orthogonality diagnostic)**.** Give a rigorous worked analysis of Orthogonality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Distinct sine and cosine modes are orthogonal on a full period. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 9.6 (Fourier coefficients diagnostic)**.** Give a rigorous worked analysis of Fourier coefficients in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Coefficients are obtained by integrating against the corresponding orthogonal modes. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 9.7 (Partial sums diagnostic)**.** Give a rigorous worked analysis of Partial sums in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The Nth Fourier partial sum is the orthogonal projection onto trigonometric polynomials of degree at most N. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

9.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 9.8 (Periodic functions diagnostic)**.** Give a rigorous worked analysis of Periodic functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A 2π-periodic function can be viewed as a function on the circle, making trigonometric modes natural coordinates. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 9.9 (Orthogonality diagnostic)**.** Give a rigorous worked analysis of Orthogonality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Distinct sine and cosine modes are orthogonal on a full period. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 9.10 (Fourier coefficients diagnostic)**.** Give a rigorous worked analysis of Fourier coefficients in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Coefficients are obtained by integrating against the corresponding orthogonal modes. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 9.11 (Partial sums diagnostic)**.** Give a rigorous worked analysis of Partial sums in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The Nth Fourier partial sum is the orthogonal projection onto trigonometric polynomials of degree at most N. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 9.12 (Bessel inequality diagnostic)**.** Give a rigorous worked analysis of Bessel inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Finite coefficient energy cannot exceed the L^2 energy of the function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 9.13 (Dirichlet kernel diagnostic)**.** Give a rigorous worked analysis of Dirichlet kernel in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Partial sums can be written as convolution with the Dirichlet kernel, making convergence an approximate-identity problem with oscillatory kernels. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 9.14 (Pointwise convergence diagnostic)**.** Give a rigorous worked analysis of Pointwise convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Piecewise smooth functions converge to their value at continuity points and to midpoint values at jumps under classical Dirichlet hypotheses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 9.15 (Gibbs phenomenon diagnostic)**.** Give a rigorous worked analysis of Gibbs phenomenon in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Near jumps, overshoot persists in relative height while its spatial width shrinks. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 9.16 (Orthogonality relations proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: Distinct trigonometric modes are orthogonal in $L^2[-\pi, \pi]$.

Solution. Use product-to-sum identities; every nonconstant integer-frequency sine or cosine integrates to zero over a full period. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 9.17 (Bessel inequality proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: The square-sum of Fourier coefficients over any finite set of orthonormal modes is bounded by $\|f\|_2^2$.

Solution. Subtract the finite orthogonal projection from f and expand the nonnegative squared norm using orthogonality. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 9.18 (Dirichlet convergence principle proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: Under standard piecewise smooth hypotheses, Fourier partial sums converge at x to $(f(x^-) + f(x^+))/2$.

Solution. Express the partial sum by the Dirichlet kernel, split the integral into a small neighborhood and its complement, use local one-sided regularity near zero and oscillatory cancellation away from zero. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 9.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Fourier series decompose periodic functions into orthogonal trigonometric modes. Coefficients are projections, partial sums are finite-dimensional approximations, and convergence depends on both function regularity and the geometry of the trigonometric system. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

9.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 9.20. State and apply the central principle behind Periodic functions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A 2π-periodic function can be viewed as a function on the circle, making trigonometric modes natural coordinates. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 9.21. State and apply the central principle behind Orthogonality. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Distinct sine and cosine modes are orthogonal on a full period. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 9.22. State and apply the central principle behind Fourier coefficients. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Coefficients are obtained by integrating against the corresponding orthogonal modes. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 9.23. State and apply the central principle behind Partial sums. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The Nth Fourier partial sum is the orthogonal projection onto trigonometric polynomials of degree at most N. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 9.24. State and apply the central principle behind Bessel inequality. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Finite coefficient energy cannot exceed the L^2 energy of the function. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 9.25. State and apply the central principle behind Dirichlet kernel. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Partial sums can be written as convolution with the Dirichlet kernel, making convergence an approximate-identity problem with oscillatory kernels. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 9.26. State and apply the central principle behind Pointwise convergence. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Piecewise smooth functions converge to their value at continuity points and to midpoint values at jumps under classical Dirichlet hypotheses. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 9.27. State and apply the central principle behind Gibbs phenomenon. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Near jumps, overshoot persists in relative height while its spatial width shrinks. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 10

Convolution and Approximate Identities

Convolution averages one function against translates of another. It interacts cleanly with translation, integration, differentiation, and Fourier transformation, and approximate identities use it to smooth and recover functions.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

10.1 Definition of convolution

For suitable functions, $(f * g)(x) = \int f(x-y)g(y)dy$.

10.2 Commutativity and associativity

Under absolute-integrability hypotheses, changes of variables and Fubini prove $f * g = g * f$ and associativity.

10.3 Young inequalities

Convolution maps certain L^p pairs into controlled L^r spaces.

10.4 Translation viewpoint

Convolution is an average of translates of one factor weighted by the other.

10.5 Approximate identities

Kernels concentrating mass near zero recover functions in norm or pointwise under appropriate hypotheses.

10.6 Smoothing

Convolution with a smooth compactly supported kernel produces smooth functions.

10.7 Differentiation

Derivatives pass to the smooth kernel: $D^\alpha(f * \phi) = f * D^\alpha\phi$ under standard assumptions.

10.8 Support

For compact supports, $\mathrm{supp}(f * g)$ lies in the Minkowski sum of the two supports.

10.9 Core structural results

Theorem 10.1 (Commutativity)**.** *If the convolution integral is absolutely meaningful, then* $f * g = g * f$*.*

Proof. Substitute $z = x - y$ in the defining integral. □

Theorem 10.2 (Young L^1 inequality)**.** *If* $f, g \in L^1(\mathbb{R}^n)$*, then* $f * g \in L^1$ *and* $\|f * g\|_1 \leq \|f\|_1\|g\|_1$*.*

Proof. Use Tonelli on $|f(x - y)g(y)|$ and translation invariance of Lebesgue measure. □

Theorem 10.3 (Approximate-identity recovery)**.** *If* ϕ_ε *is an approximate identity and* $f \in L^p$*,* $1 \leq p < \infty$*, then* $f * \phi_\varepsilon \to f$ *in* L^p *under standard normalization.*

Proof. Write the difference as an average of translations $\tau_y f - f$, use continuity of translations in L^p, concentrate kernel mass near zero, and control the far tail by kernel mass. □

10.10 Worked examples

Example 10.4 (Definition of convolution diagnostic)**.** Give a rigorous worked analysis of Definition of convolution in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For suitable functions, $(f * g)(x) = \int f(x - y)g(y)dy$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 10.5 (Commutativity and associativity diagnostic)**.** Give a rigorous worked analysis of Commutativity and associativity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Under absolute-integrability hypotheses, changes of variables and Fubini prove $f * g = g * f$ and associativity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 10.6 (Young inequalities diagnostic)**.** Give a rigorous worked analysis of Young inequalities in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Convolution maps certain L^p pairs into controlled L^r spaces. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 10.7 (Translation viewpoint diagnostic)**.** Give a rigorous worked analysis of Translation viewpoint in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Convolution is an average of translates of one factor weighted by the other. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

10.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 10.8 (Definition of convolution diagnostic)**.** Give a rigorous worked analysis of Definition of convolution in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For suitable functions, $(f * g)(x) = \int f(x-y)g(y)dy$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 10.9 (Commutativity and associativity diagnostic)**.** Give a rigorous worked analysis of Commutativity and associativity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Under absolute-integrability hypotheses, changes of variables and Fubini prove $f*g = g*f$ and associativity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 10.10 (Young inequalities diagnostic)**.** Give a rigorous worked analysis of Young inequalities in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Convolution maps certain L^p pairs into controlled L^r spaces. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 10.11 (Translation viewpoint diagnostic)**.** Give a rigorous worked analysis of Translation viewpoint in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Convolution is an average of translates of one factor weighted by the other. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 10.12 (Approximate identities diagnostic)**.** Give a rigorous worked analysis of Approximate identities in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Kernels concentrating mass near zero recover functions in norm or pointwise under appropriate hypotheses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 10.13 (Smoothing diagnostic)**.** Give a rigorous worked analysis of Smoothing in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Convolution with a smooth compactly supported kernel produces smooth functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 10.14 (Differentiation diagnostic)**.** Give a rigorous worked analysis of Differentiation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Derivatives pass to the smooth kernel: $D^\alpha(f * \phi) = f * D^\alpha \phi$ under standard assumptions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 10.15 (Support diagnostic)**.** Give a rigorous worked analysis of Support in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For compact supports, $\operatorname{supp}(f * g)$ lies in the Minkowski sum of the two supports. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 10.16 (Commutativity proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If the convolution integral is absolutely meaningful, then $f * g = g * f$.

Solution. Substitute $z = x - y$ in the defining integral. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 10.17 (Young L^1 inequality proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f, g \in L^1(\mathbb{R}^n)$, then $f * g \in L^1$ and $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$.

Solution. Use Tonelli on $|f(x-y)g(y)|$ and translation invariance of Lebesgue measure. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 10.18 (Approximate-identity recovery proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If ϕ_ε is an approximate identity and $f \in L^p$, $1 \leq p < \infty$, then $f * \phi_\varepsilon \to f$ in L^p under standard normalization.

Solution. Write the difference as an average of translations $\tau_y f - f$, use continuity of translations in L^p, concentrate kernel mass near zero, and control the far tail by kernel mass. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 10.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Convolution averages one function against translates of another. It interacts cleanly with translation, integration, differentiation, and Fourier transformation, and approximate identities use it to smooth and recover functions. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

10.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 10.20. State and apply the central principle behind Definition of convolution. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. For suitable functions, $(f*g)(x)=\int f(x-y)g(y)dy$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 10.21. State and apply the central principle behind Commutativity and associativity. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Under absolute-integrability hypotheses, changes of variables and Fubini prove $f*g=g*f$ and associativity. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 10.22. State and apply the central principle behind Young inequalities. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Convolution maps certain L^p pairs into controlled L^r spaces. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 10.23. State and apply the central principle behind Translation viewpoint. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Convolution is an average of translates of one factor weighted by the other. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 10.24. State and apply the central principle behind Approximate identities. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Kernels concentrating mass near zero recover functions in norm or pointwise under appropriate hypotheses. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 10.25. State and apply the central principle behind Smoothing. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Convolution with a smooth compactly supported kernel produces smooth functions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 10.26. State and apply the central principle behind Differentiation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Derivatives pass to the smooth kernel: $D^\alpha(f * \phi) = f * D^\alpha \phi$ under standard assumptions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 10.27. State and apply the central principle behind Support. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. For compact supports, $\mathrm{supp}(f * g)$ lies in the Minkowski sum of the two supports. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 11

The Fourier Transform

The Fourier transform converts translation and differentiation into multiplication, convolution into products, and spatial localization into frequency information. It is the central transform chapter of Volume III and is developed with extra structural depth.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

11.1 Definition on L^1

Fix the convention $\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i x\cdot\xi}dx$ for integrable functions.

11.2 Boundedness and continuity

The transform of an L^1 function is bounded and uniformly continuous, with $\|\widehat{f}\|_\infty \leq \|f\|_1$.

11.3 Riemann–Lebesgue lemma

Fourier transforms of L^1 functions vanish at infinity.

11.4 Translation and modulation

Translation in physical space creates a phase factor in frequency; modulation shifts the transform.

11.5 Scaling

Dilation in space inversely dilates frequency and contributes the Jacobian factor.

11.6 Differentiation and moments

Differentiation becomes multiplication by $2\pi i\xi$, while multiplication by x becomes differentiation in frequency.

11.7 Convolution theorem

Convolution transforms into pointwise multiplication, and products transform into convolution when hypotheses permit.

11.8 Inversion

Under suitable integrability or Schwartz hypotheses, applying the inverse transform reconstructs the original function.

11.9 Symmetry and reality

Even, odd, real, and radial structures impose corresponding symmetry on the transform.

11.10 Uncertainty viewpoint

Strong concentration in space conflicts with strong concentration in frequency, foreshadowing quantitative uncertainty principles.

11.11 Core structural results

Theorem 11.1 (L^1 bound)**.** *For $f \in L^1$, $|\widehat{f}(\xi)| \le \|f\|_1$ for every ξ.*

Proof. Take absolute values inside the integral; the complex exponential has modulus one. □

Theorem 11.2 (Riemann–Lebesgue lemma)**.** *If $f \in L^1(\mathbb{R}^n)$, then $\widehat{f}(\xi) \to 0$ as $|\xi| \to \infty$.*

Proof. First prove it for smooth compactly supported functions by integration by parts or oscillatory cancellation, then approximate an arbitrary L^1 function in L^1 and use the uniform transform bound. □

Theorem 11.3 (Convolution theorem)**.** *If $f, g \in L^1$, then $\widehat{f * g} = \widehat{f}\,\widehat{g}$.*

Proof. Insert the convolution integral, use Fubini justified by absolute integrability, change variables $u = x - y$, and factor the two resulting integrals. □

Theorem 11.4 (Fourier inversion on Schwartz space)**.** *For $f \in \mathcal{S}(\mathbb{R}^n)$, inverse transformation recovers f.*

Proof. Regularize with a Gaussian multiplier, use Fubini to identify the regularized inverse as convolution with an approximate identity, and pass to the limit in the Schwartz or uniform topology. □

11.12 Worked examples

Example 11.5 (Definition on L^1 diagnostic)**.** Give a rigorous worked analysis of Definition on L^1 in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Fix the convention $\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i x\cdot\xi}dx$ for integrable functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 11.6 (Boundedness and continuity diagnostic)**.** Give a rigorous worked analysis of Boundedness and continuity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The transform of an L^1 function is bounded and uniformly continuous, with $\|\widehat{f}\|_\infty \le \|f\|_1$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 11.7 (Riemann–Lebesgue lemma diagnostic)**.** Give a rigorous worked analysis of Riemann–Lebesgue lemma in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Fourier transforms of L^1 functions vanish at infinity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 11.8 (Translation and modulation diagnostic)**.** Give a rigorous worked analysis of Translation and modulation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Translation in physical space creates a phase factor in frequency; modulation shifts the transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

11.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 11.9 (Definition on L^1 diagnostic)**.** Give a rigorous worked analysis of Definition on L^1 in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Fix the convention $\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i x\cdot\xi}dx$ for integrable functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 11.10 (Boundedness and continuity diagnostic)**.** Give a rigorous worked analysis of Boundedness and continuity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The transform of an L^1 function is bounded and uniformly continuous, with $\|\widehat{f}\|_\infty \le \|f\|_1$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 11.11 (Riemann–Lebesgue lemma diagnostic)**.** Give a rigorous worked analysis of Riemann–Lebesgue lemma in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Fourier transforms of L^1 functions vanish at infinity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 11.12 (Translation and modulation diagnostic)**.** Give a rigorous worked analysis of Translation and modulation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Translation in physical space creates a phase factor in frequency; modulation shifts the transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 11.13 (Scaling diagnostic)**.** Give a rigorous worked analysis of Scaling in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Dilation in space inversely dilates frequency and contributes the Jacobian factor. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 11.14 (Differentiation and moments diagnostic)**.** Give a rigorous worked analysis of Differentiation and moments in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Differentiation becomes multiplication by $2\pi i\xi$, while multiplication by x becomes differentiation in frequency. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 11.15 (Convolution theorem diagnostic)**.** Give a rigorous worked analysis of Convolution theorem in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Convolution transforms into pointwise multiplication, and products transform into convolution when hypotheses permit. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 11.16 (L^1 bound proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For $f \in L^1$, $|\widehat{f}(\xi)| \le \|f\|_1$ for every ξ.

Solution. Take absolute values inside the integral; the complex exponential has modulus one. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 11.17 (Riemann–Lebesgue lemma proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in L^1(\mathbb{R}^n)$, then $\widehat{f}(\xi) \to 0$ as $|\xi| \to \infty$.

Solution. First prove it for smooth compactly supported functions by integration by parts or oscillatory cancellation, then approximate an arbitrary L^1 function in L^1 and use the uniform transform bound. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 11.18 (Convolution theorem proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f, g \in L^1$, then $\widehat{f * g} = \widehat{f}\,\widehat{g}$.

Solution. Insert the convolution integral, use Fubini justified by absolute integrability, change variables $u = x - y$, and factor the two resulting integrals. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 11.19 (Fourier inversion on Schwartz space proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For $f \in \mathcal{S}(\mathbb{R}^n)$, inverse transformation recovers f.

Solution. Regularize with a Gaussian multiplier, use Fubini to identify the regularized inverse as convolution with an approximate identity, and pass to the limit in the Schwartz or uniform topology. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 11.20 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. The Fourier transform converts translation and differentiation into multiplication, convolution into products, and spatial localization into frequency information. It is the central transform chapter of Volume III and is developed with extra structural depth. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

11.14 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 11.21. State and apply the central principle behind Definition on L^1. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Fix the convention $\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i x\cdot\xi}dx$ for integrable functions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 11.22. State and apply the central principle behind Boundedness and continuity. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The transform of an L^1 function is bounded and uniformly continuous, with $\|\widehat{f}\|_\infty \le \|f\|_1$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 11.23. State and apply the central principle behind Riemann–Lebesgue lemma. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Fourier transforms of L^1 functions vanish at infinity. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 11.24. State and apply the central principle behind Translation and modulation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Translation in physical space creates a phase factor in frequency; modulation shifts the transform. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 11.25. State and apply the central principle behind Scaling. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Dilation in space inversely dilates frequency and contributes the Jacobian factor. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 11.26. State and apply the central principle behind Differentiation and moments. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Differentiation becomes multiplication by $2\pi i\xi$, while multiplication by x becomes differentiation in frequency. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 11.27. State and apply the central principle behind Convolution theorem. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Convolution transforms into pointwise multiplication, and products transform into convolution when hypotheses permit. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 11.28. State and apply the central principle behind Inversion. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Under suitable integrability or Schwartz hypotheses, applying the inverse transform reconstructs the original function. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 12

The Gaussian and Transform Calculus

The Gaussian is the model function of transform calculus: it is smooth, rapidly decaying, and mapped to another Gaussian. Differentiation, scaling, and tensor-product arguments then generate a large transform table.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

12.1 Gaussian integrals

The integral of $e^{-\pi x^2}$ over the line is one under the standard normalization.

12.2 Self-reciprocity

With the 2π Fourier convention, $e^{-\pi|x|^2}$ is its own Fourier transform.

12.3 Scaling

Scaled Gaussians transform to reciprocal-width Gaussians with the appropriate determinant factor.

12.4 Differentiated Gaussians

Polynomial times Gaussian functions arise by differentiating in space or frequency.

12.5 Heat kernels

Gaussian kernels are the fundamental averaging kernels of the heat equation.

12.6 Tensor products

The n-dimensional transform of a product of one-dimensional Gaussians factors coordinatewise.

12.7 Hermite structure

Repeated differentiation of Gaussians produces Hermite-polynomial factors and a transform-invariant basis structure.

12.8 Transform tables

Translation, modulation, scaling, differentiation, and convolution generate many explicit transform pairs from a few seeds.

12.9 Core structural results

Theorem 12.1 (Gaussian transform). *Under the convention $\widehat{f}(\xi) = \int f(x)e^{-2\pi i x\xi}dx$, $e^{-\pi x^2}$ transforms to $e^{-\pi\xi^2}$.*

Proof. Differentiate the transform with respect to ξ, integrate by parts to obtain an ODE, solve the ODE, and determine the constant from the Gaussian integral at zero. □

Theorem 12.2 (Scaling rule). *If $f_a(x) = f(ax)$ with $a \neq 0$, then $\widehat{f_a}(\xi) = |a|^{-1}\widehat{f}(\xi/a)$ in one dimension.*

Proof. Substitute $u = ax$ in the transform integral and account for the Jacobian. □

Theorem 12.3 (Heat-kernel semigroup). *Gaussian heat kernels satisfy $G_t * G_s = G_{t+s}$.*

Proof. Fourier transformation turns convolution into multiplication, while $\widehat{G}_t$ is an exponential multiplier whose product at times t, s equals the multiplier at $t + s$; invert the transform. □

12.10 Worked examples

Example 12.4 (Gaussian integrals diagnostic). Give a rigorous worked analysis of Gaussian integrals in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The integral of $e^{-\pi x^2}$ over the line is one under the standard normalization. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 12.5 (Self-reciprocity diagnostic). Give a rigorous worked analysis of Self-reciprocity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. With the 2π Fourier convention, $e^{-\pi|x|^2}$ is its own Fourier transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 12.6 (Scaling diagnostic). Give a rigorous worked analysis of Scaling in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Scaled Gaussians transform to reciprocal-width Gaussians with the appropriate determinant factor. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 12.7 (Differentiated Gaussians diagnostic)**.** Give a rigorous worked analysis of Differentiated Gaussians in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Polynomial times Gaussian functions arise by differentiating in space or frequency. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

12.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 12.8 (Gaussian integrals diagnostic)**.** Give a rigorous worked analysis of Gaussian integrals in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The integral of $e^{-\pi x^2}$ over the line is one under the standard normalization. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 12.9 (Self-reciprocity diagnostic)**.** Give a rigorous worked analysis of Self-reciprocity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. With the 2π Fourier convention, $e^{-\pi|x|^2}$ is its own Fourier transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 12.10 (Scaling diagnostic)**.** Give a rigorous worked analysis of Scaling in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Scaled Gaussians transform to reciprocal-width Gaussians with the appropriate determinant factor. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 12.11 (Differentiated Gaussians diagnostic)**.** Give a rigorous worked analysis of Differentiated Gaussians in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Polynomial times Gaussian functions arise by differentiating in space or frequency. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 12.12 (Heat kernels diagnostic)**.** Give a rigorous worked analysis of Heat kernels in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Gaussian kernels are the fundamental averaging kernels of the heat equation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 12.13 (Tensor products diagnostic)**.** Give a rigorous worked analysis of Tensor products in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The n-dimensional transform of a product of one-dimensional Gaussians factors coordinatewise. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 12.14 (Hermite structure diagnostic)**.** Give a rigorous worked analysis of Hermite structure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Repeated differentiation of Gaussians produces Hermite-polynomial factors and a transform-invariant basis structure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 12.15 (Transform tables diagnostic)**.** Give a rigorous worked analysis of Transform tables in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Translation, modulation, scaling, differentiation, and convolution generate many explicit transform pairs from a few seeds. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 12.16 (Gaussian transform proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: Under the convention $\widehat{f}(\xi) = \int f(x)e^{-2\pi i x\xi}dx$, $e^{-\pi x^2}$ transforms to $e^{-\pi\xi^2}$.

Solution. Differentiate the transform with respect to ξ, integrate by parts to obtain an ODE, solve the ODE, and determine the constant from the Gaussian integral at zero. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 12.17 (Scaling rule proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f_a(x) = f(ax)$ with $a \neq 0$, then $\widehat{f_a}(\xi) = |a|^{-1}\widehat{f}(\xi/a)$ in one dimension.

Solution. Substitute $u = ax$ in the transform integral and account for the Jacobian. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 12.18 (Heat-kernel semigroup proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: Gaussian heat kernels satisfy $G_t * G_s = G_{t+s}$.

Solution. Fourier transformation turns convolution into multiplication, while $\widehat{G}_t$ is an exponential multiplier whose product at times t, s equals the multiplier at $t + s$; invert the transform. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 12.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. The Gaussian is the model function of transform calculus: it is smooth, rapidly decaying, and mapped to another Gaussian. Differentiation, scaling, and tensor-product arguments then generate a large transform table. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

12.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 12.20. State and apply the central principle behind Gaussian integrals. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. The integral of $e^{-\pi x^2}$ over the line is one under the standard normalization. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 12.21. State and apply the central principle behind Self-reciprocity. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. With the 2π Fourier convention, $e^{-\pi|x|^2}$ is its own Fourier transform. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 12.22. State and apply the central principle behind Scaling. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Scaled Gaussians transform to reciprocal-width Gaussians with the appropriate determinant factor. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 12.23. State and apply the central principle behind Differentiated Gaussians. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Polynomial times Gaussian functions arise by differentiating in space or frequency. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 12.24. State and apply the central principle behind Heat kernels. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Gaussian kernels are the fundamental averaging kernels of the heat equation. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 12.25. State and apply the central principle behind Tensor products. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. The n-dimensional transform of a product of one-dimensional Gaussians factors coordinatewise. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 12.26. State and apply the central principle behind Hermite structure. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Repeated differentiation of Gaussians produces Hermite-polynomial factors and a transform-invariant basis structure. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 12.27. State and apply the central principle behind Transform tables. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Translation, modulation, scaling, differentiation, and convolution generate many explicit transform pairs from a few seeds. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 13

Plancherel and L2 Fourier Theory

Plancherel theory extends the Fourier transform from integrable functions to a unitary operator on L^2. Orthogonality and energy conservation become the main language, allowing transform methods beyond pointwise integral formulas.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

13.1 Parseval identity

For suitable functions, the inner product is preserved by the Fourier transform.

13.2 Plancherel extension

Density of Schwartz functions extends the transform uniquely to all of L^2.

13.3 Unitarity

The L^2 transform preserves norm and inner product.

13.4 Inverse transform in L^2

Inversion holds in the Hilbert-space sense even when neither function nor transform is integrable.

13.5 Convolution in L^2 settings

Combining L^1 and L^2 hypotheses yields useful convolution bounds and multiplier formulas.

13.6 Multipliers

Bounded frequency multipliers define bounded operators on L^2.

13.7 Energy methods

Differentiation identities translate Sobolev-type derivative energy into weighted frequency energy.

13.8 Approximation

Frequency cutoffs approximate L^2 functions and provide smooth or band-limited regularizations.

13.9 Core structural results

Theorem 13.1 (Plancherel theorem)**.** *The Fourier transform extends uniquely to a unitary operator on $L^2(\mathbb{R}^n)$.*

Proof. Prove norm preservation on Schwartz functions, use density of Schwartz space in L^2 to define the extension by completion, and preserve the norm in the limit. ☐

Theorem 13.2 (Parseval identity)**.** *For $f, g \in L^2$, $\langle \widehat{f}, \widehat{g} \rangle = \langle f, g \rangle$.*

Proof. Use polarization of the norm identity obtained from Plancherel. ☐

Theorem 13.3 (L^2 multiplier bound)**.** *If $m \in L^\infty$ and $Tf = \mathcal{F}^{-1}(m\widehat{f})$, then $\|Tf\|_2 \leq \|m\|_\infty \|f\|_2$.*

Proof. Apply Plancherel and the pointwise bound $|m\widehat{f}| \leq \|m\|_\infty |\widehat{f}|$. ☐

13.10 Worked examples

Example 13.4 (Parseval identity diagnostic)**.** Give a rigorous worked analysis of Parseval identity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For suitable functions, the inner product is preserved by the Fourier transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 13.5 (Plancherel extension diagnostic)**.** Give a rigorous worked analysis of Plancherel extension in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Density of Schwartz functions extends the transform uniquely to all of L^2. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 13.6 (Unitarity diagnostic)**.** Give a rigorous worked analysis of Unitarity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The L^2 transform preserves norm and inner product. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 13.7 (Inverse transform in L^2 diagnostic)**.** Give a rigorous worked analysis of Inverse transform in L^2 in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Inversion holds in the Hilbert-space sense even when neither function nor transform is integrable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

13.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 13.8 (Parseval identity diagnostic)**.** Give a rigorous worked analysis of Parseval identity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For suitable functions, the inner product is preserved by the Fourier transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 13.9 (Plancherel extension diagnostic)**.** Give a rigorous worked analysis of Plancherel extension in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Density of Schwartz functions extends the transform uniquely to all of L^2. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 13.10 (Unitarity diagnostic)**.** Give a rigorous worked analysis of Unitarity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The L^2 transform preserves norm and inner product. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 13.11 (Inverse transform in L^2 diagnostic)**.** Give a rigorous worked analysis of Inverse transform in L^2 in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Inversion holds in the Hilbert-space sense even when neither function nor transform is integrable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 13.12 (Convolution in L^2 settings diagnostic)**.** Give a rigorous worked analysis of Convolution in L^2 settings in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Combining L^1 and L^2 hypotheses yields useful convolution bounds and multiplier formulas. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 13.13 (Multipliers diagnostic)**.** Give a rigorous worked analysis of Multipliers in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Bounded frequency multipliers define bounded operators on L^2. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 13.14 (Energy methods diagnostic)**.** Give a rigorous worked analysis of Energy methods in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Differentiation identities translate Sobolev-type derivative energy into weighted frequency energy. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 13.15 (Approximation diagnostic)**.** Give a rigorous worked analysis of Approximation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Frequency cutoffs approximate L^2 functions and provide smooth or band-limited regularizations. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 13.16 (Plancherel theorem proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: The Fourier transform extends uniquely to a unitary operator on $L^2(\mathbb{R}^n)$.

Solution. Prove norm preservation on Schwartz functions, use density of Schwartz space in L^2 to define the extension by completion, and preserve the norm in the limit. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 13.17 (Parseval identity proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For $f, g \in L^2$, $\langle \widehat{f}, \widehat{g} \rangle = \langle f, g \rangle$.

Solution. Use polarization of the norm identity obtained from Plancherel. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 13.18 (L^2 multiplier bound proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $m \in L^\infty$ and $Tf = \mathcal{F}^{-1}(m\widehat{f})$, then $\|Tf\|_2 \le \|m\|_\infty \|f\|_2$.

Solution. Apply Plancherel and the pointwise bound $|m\widehat{f}| \le \|m\|_\infty |\widehat{f}|$. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 13.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Plancherel theory extends the Fourier transform from integrable functions to a unitary operator on L^2. Orthogonality and energy conservation become the main language, allowing transform methods beyond pointwise integral formulas. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

13.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 13.20. State and apply the central principle behind Parseval identity. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. For suitable functions, the inner product is preserved by the Fourier transform. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 13.21. State and apply the central principle behind Plancherel extension. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Density of Schwartz functions extends the transform uniquely to all of L^2. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 13.22. State and apply the central principle behind Unitarity. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The L^2 transform preserves norm and inner product. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 13.23. State and apply the central principle behind Inverse transform in L^2. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Inversion holds in the Hilbert-space sense even when neither function nor transform is integrable. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 13.24. State and apply the central principle behind Convolution in L^2 settings. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Combining L^1 and L^2 hypotheses yields useful convolution bounds and multiplier formulas. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 13.25. State and apply the central principle behind Multipliers. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Bounded frequency multipliers define bounded operators on L^2. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 13.26. State and apply the central principle behind Energy methods. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. $\square$

Solution. Differentiation identities translate Sobolev-type derivative energy into weighted frequency energy. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 13.27. State and apply the central principle behind Approximation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. $\square$

Solution. Frequency cutoffs approximate L^2 functions and provide smooth or band-limited regularizations. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 14

The Schwartz Space

The Schwartz space packages smoothness and rapid decay into a Fréchet-type test-function space stable under Fourier transformation. It is the natural common domain for classical transform calculus and tempered distributions.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

14.1 Rapid decay

A Schwartz function decays faster than every polynomial together with all derivatives.

14.2 Schwartz seminorms

Quantities $\sup_x |x^\alpha D^\beta f(x)|$ encode the topology.

14.3 Algebraic stability

Multiplication by polynomials and differentiation preserve Schwartz space.

14.4 Translation and modulation

Translations and frequency modulations preserve the class.

14.5 Fourier invariance

The Fourier transform maps Schwartz space bijectively to itself.

14.6 Convolution

Schwartz functions convolve to Schwartz functions; convolution with appropriate integrable functions smooths while retaining decay.

14.7 Density

Schwartz space is dense in L^p for finite p and in many Sobolev spaces.

14.8 Role in distribution theory

Its rapid decay makes pairing with polynomial-growth objects well-defined, leading to tempered distributions.

14.9 Core structural results

Theorem 14.1 (Fourier invariance). *If $f \in \mathcal{S}$, then $\widehat{f} \in \mathcal{S}$.*

Proof. Differentiation in frequency corresponds to multiplication by powers of x, while multiplication by powers of ξ corresponds to derivatives of f; all such weighted derivatives are bounded and integrable for Schwartz functions. ☐

Theorem 14.2 (Algebraic stability). *If $f \in \mathcal{S}$, then $x^\alpha D^\beta f \in \mathcal{S}$.*

Proof. Every Schwartz seminorm of this expression is a finite linear combination of higher Schwartz seminorms of f. ☐

Theorem 14.3 (Schwartz convolution closure). *If $f, g \in \mathcal{S}$, then $f * g \in \mathcal{S}$.*

Proof. Differentiate under the integral and distribute polynomial weights between $x - y$ and y; rapid decay of both factors controls every resulting seminorm. ☐

14.10 Worked examples

Example 14.4 (Rapid decay diagnostic). Give a rigorous worked analysis of Rapid decay in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A Schwartz function decays faster than every polynomial together with all derivatives. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 14.5 (Schwartz seminorms diagnostic). Give a rigorous worked analysis of Schwartz seminorms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Quantities $\sup_x |x^\alpha D^\beta f(x)|$ encode the topology. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 14.6 (Algebraic stability diagnostic). Give a rigorous worked analysis of Algebraic stability in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Multiplication by polynomials and differentiation preserve Schwartz space. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 14.7 (Translation and modulation diagnostic)**.** Give a rigorous worked analysis of Translation and modulation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Translations and frequency modulations preserve the class. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

14.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 14.8 (Rapid decay diagnostic)**.** Give a rigorous worked analysis of Rapid decay in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A Schwartz function decays faster than every polynomial together with all derivatives. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 14.9 (Schwartz seminorms diagnostic)**.** Give a rigorous worked analysis of Schwartz seminorms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Quantities $\sup_x |x^\alpha D^\beta f(x)|$ encode the topology. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 14.10 (Algebraic stability diagnostic)**.** Give a rigorous worked analysis of Algebraic stability in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Multiplication by polynomials and differentiation preserve Schwartz space. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 14.11 (Translation and modulation diagnostic)**.** Give a rigorous worked analysis of Translation and modulation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Translations and frequency modulations preserve the class. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 14.12 (Fourier invariance diagnostic)**.** Give a rigorous worked analysis of Fourier invariance in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The Fourier transform maps Schwartz space bijectively to itself. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 14.13 (Convolution diagnostic)**.** Give a rigorous worked analysis of Convolution in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Schwartz functions convolve to Schwartz functions; convolution with appropriate integrable functions smooths while retaining decay. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 14.14 (Density diagnostic)**.** Give a rigorous worked analysis of Density in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Schwartz space is dense in L^p for finite p and in many Sobolev spaces. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 14.15 (Role in distribution theory diagnostic)**.** Give a rigorous worked analysis of Role in distribution theory in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Its rapid decay makes pairing with polynomial-growth objects well-defined, leading to tempered distributions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 14.16 (Fourier invariance proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in \mathcal{S}$, then $\widehat{f} \in \mathcal{S}$.

Solution. Differentiation in frequency corresponds to multiplication by powers of x, while multiplication by powers of ξ corresponds to derivatives of f; all such weighted derivatives are bounded and integrable for Schwartz functions. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 14.17 (Algebraic stability proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in \mathcal{S}$, then $x^\alpha D^\beta f \in \mathcal{S}$.

Solution. Every Schwartz seminorm of this expression is a finite linear combination of higher Schwartz seminorms of f. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 14.18 (Schwartz convolution closure proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f, g \in \mathcal{S}$, then $f * g \in \mathcal{S}$.

Solution. Differentiate under the integral and distribute polynomial weights between $x - y$ and y; rapid decay of both factors controls every resulting seminorm. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 14.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. The Schwartz space packages smoothness and rapid decay into a Fréchet-type test-function space stable under Fourier transformation. It is the natural common domain for classical transform calculus and tempered distributions. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

14.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 14.20. State and apply the central principle behind Rapid decay. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A Schwartz function decays faster than every polynomial together with all derivatives. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 14.21. State and apply the central principle behind Schwartz seminorms. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Quantities $\sup_x |x^\alpha D^\beta f(x)|$ encode the topology. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 14.22. State and apply the central principle behind Algebraic stability. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Multiplication by polynomials and differentiation preserve Schwartz space. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 14.23. State and apply the central principle behind Translation and modulation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Translations and frequency modulations preserve the class. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 14.24. State and apply the central principle behind Fourier invariance. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The Fourier transform maps Schwartz space bijectively to itself. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 14.25. State and apply the central principle behind Convolution. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Schwartz functions convolve to Schwartz functions; convolution with appropriate integrable functions smooths while retaining decay. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 14.26. State and apply the central principle behind Density. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Schwartz space is dense in L^p for finite p and in many Sobolev spaces. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 14.27. State and apply the central principle behind Role in distribution theory. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Its rapid decay makes pairing with polynomial-growth objects well-defined, leading to tempered distributions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Part III

Distribution Theory

Chapter 15

Test-Function Spaces

Test-function spaces provide compactly supported smooth probes. Their topology is stronger and more local than the Schwartz topology, and distributions are defined as continuous linear functionals on these probes.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

15.1 Smooth compact support

$C_c^\infty(\Omega)$ consists of infinitely differentiable functions whose supports are compact subsets of the open set Ω.

15.2 Bump functions

Smooth cutoffs allow localization without sharp discontinuities.

15.3 Support control

A sequence converging in the test-function topology eventually has support inside one fixed compact set.

15.4 Derivative control

Within that compact set, every derivative converges uniformly.

15.5 Partitions of unity

Locally finite smooth partitions subordinate to open covers allow local constructions to be patched globally.

15.6 Localization

Multiplying by a test function isolates distributional behavior in a chosen region.

15.7 Comparison with Schwartz space

Compact support implies rapid decay after extension by zero, but the topologies and intended duals differ.

15.8 Dual space preview

The continuous dual of $C_c^\infty(\Omega)$ is the space of distributions on Ω.

15.9 Core structural results

Theorem 15.1 (Existence of bump functions)**.** *Given compact $K \subset U$ with U open in $\mathbb{R}^n$, there is $\phi \in C_c^\infty(U)$ with $0 \le \phi \le 1$ and $\phi = 1$ near K.*

Proof. Convolve the indicator of a slightly enlarged neighborhood of K with a sufficiently small smooth mollifier, then truncate the geometric construction so the support stays inside U. □

Theorem 15.2 (Partition of unity)**.** *Every open cover of a paracompact smooth Euclidean domain admits a smooth locally finite partition of unity subordinate to it.*

Proof. Refine the cover by locally finite relatively compact sets, choose bump functions on the refinement, and normalize their locally finite positive sum. □

Theorem 15.3 (Test-function convergence)**.** *If $\phi_j \to 0$ in $C_c^\infty(\Omega)$, then all supports eventually lie in one compact $K \Subset \Omega$ and every derivative converges uniformly on K.*

Proof. This is the defining sequential behavior of the standard LF topology and is precisely the control needed for distributional continuity. □

15.10 Worked examples

Example 15.4 (Smooth compact support diagnostic)**.** Give a rigorous worked analysis of Smooth compact support in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. $C_c^\infty(\Omega)$ consists of infinitely differentiable functions whose supports are compact subsets of the open set Ω. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 15.5 (Bump functions diagnostic)**.** Give a rigorous worked analysis of Bump functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Smooth cutoffs allow localization without sharp discontinuities. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 15.6 (Support control diagnostic)**.** Give a rigorous worked analysis of Support control in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A sequence converging in the test-function topology eventually has support inside one fixed compact set. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 15.7 (Derivative control diagnostic)**.** Give a rigorous worked analysis of Derivative control in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Within that compact set, every derivative converges uniformly. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

15.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 15.8 (Smooth compact support diagnostic)**.** Give a rigorous worked analysis of Smooth compact support in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. $C_c^\infty(\Omega)$ consists of infinitely differentiable functions whose supports are compact subsets of the open set Ω. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 15.9 (Bump functions diagnostic)**.** Give a rigorous worked analysis of Bump functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Smooth cutoffs allow localization without sharp discontinuities. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 15.10 (Support control diagnostic)**.** Give a rigorous worked analysis of Support control in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A sequence converging in the test-function topology eventually has support inside one fixed compact set. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 15.11 (Derivative control diagnostic)**.** Give a rigorous worked analysis of Derivative control in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Within that compact set, every derivative converges uniformly. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 15.12 (Partitions of unity diagnostic)**.** Give a rigorous worked analysis of Partitions of unity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Locally finite smooth partitions subordinate to open covers allow local constructions to be patched globally. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 15.13 (Localization diagnostic)**.** Give a rigorous worked analysis of Localization in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Multiplying by a test function isolates distributional behavior in a chosen region. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 15.14 (Comparison with Schwartz space diagnostic)**.** Give a rigorous worked analysis of Comparison with Schwartz space in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Compact support implies rapid decay after extension by zero, but the topologies and intended duals differ. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 15.15 (Dual space preview diagnostic)**.** Give a rigorous worked analysis of Dual space preview in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The continuous dual of $C_c^\infty(\Omega)$ is the space of distributions on Ω. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 15.16 (Existence of bump functions proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: Given compact $K \subset U$ with U open in $\mathbb{R}^n$, there is $\phi \in C_c^\infty(U)$ with $0 \le \phi \le 1$ and $\phi = 1$ near K.

Solution. Convolve the indicator of a slightly enlarged neighborhood of K with a sufficiently small smooth mollifier, then truncate the geometric construction so the support stays inside U. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 15.17 (Partition of unity proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: Every open cover of a paracompact smooth Euclidean domain admits a smooth locally finite partition of unity subordinate to it.

Solution. Refine the cover by locally finite relatively compact sets, choose bump functions on the refinement, and normalize their locally finite positive sum. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 15.18 (Test-function convergence proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $\phi_j \to 0$ in $C_c^\infty(\Omega)$, then all supports eventually lie in one compact $K \Subset \Omega$ and every derivative converges uniformly on K.

Solution. This is the defining sequential behavior of the standard LF topology and is precisely the control needed for distributional continuity. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 15.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Test-function spaces provide compactly supported smooth probes. Their topology is stronger and more local than the Schwartz topology, and distributions are defined as continuous linear functionals on these probes. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

15.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 15.20. State and apply the central principle behind Smooth compact support. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. $C_c^\infty(\Omega)$ consists of infinitely differentiable functions whose supports are compact subsets of the open set Ω. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 15.21. State and apply the central principle behind Bump functions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Smooth cutoffs allow localization without sharp discontinuities. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 15.22. State and apply the central principle behind Support control. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A sequence converging in the test-function topology eventually has support inside one fixed compact set. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 15.23. State and apply the central principle behind Derivative control. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Within that compact set, every derivative converges uniformly. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 15.24. State and apply the central principle behind Partitions of unity. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Locally finite smooth partitions subordinate to open covers allow local constructions to be patched globally. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 15.25. State and apply the central principle behind Localization. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Multiplying by a test function isolates distributional behavior in a chosen region. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 15.26. State and apply the central principle behind Comparison with Schwartz space. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Compact support implies rapid decay after extension by zero, but the topologies and intended duals differ. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 15.27. State and apply the central principle behind Dual space preview. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The continuous dual of $C_c^\infty(\Omega)$ is the space of distributions on Ω. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 16

Distributions and Distributional Derivatives

Distributions extend the derivative and transform calculus to nonsmooth objects by duality. A distribution acts continuously and linearly on test functions, and derivatives are transferred onto the test function by integration by parts.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

16.1 Distributions

A distribution on Ω is a continuous linear functional on $C_c^\infty(\Omega)$.

16.2 Regular distributions

Locally integrable functions define distributions by integration against test functions.

16.3 Dirac mass

Point evaluation $\delta_{x_0}(\phi) = \phi(x_0)$ is a distribution not represented by an ordinary locally integrable function.

16.4 Distributional derivatives

Define $\partial^\alpha T(\phi) = (-1)^{|\alpha|} T(\partial^\alpha \phi)$.

16.5 Integration by parts

For smooth functions, distributional differentiation agrees with classical differentiation because test functions have compact support.

16.6 Convergence of distributions

A sequence converges distributionally when it converges on every test function.

16.7 Multiplication by smooth functions

If $a \in C^\infty$, define $(aT)(\phi) = T(a\phi)$.

16.8 Linear differential operators

Smooth-coefficient differential operators act naturally on distributions by combining multiplication and distributional differentiation.

16.9 Core structural results

Theorem 16.1 (Agreement with classical derivatives)**.** *If $f \in C^1$ defines a regular distribution, then its distributional derivative is the regular distribution defined by f'.*

Proof. For every test function, $D_f'(\phi) = -\int f\phi' = \int f'\phi$ by integration by parts and vanishing boundary terms. ☐

Theorem 16.2 (Derivative of Heaviside)**.** *The distributional derivative of the Heaviside step is δ_0.*

Proof. For a test function ϕ, $H'(\phi) = -\int_0^\infty \phi'(x)dx = \phi(0)$. ☐

Theorem 16.3 (Continuity of differentiation)**.** *If $T_n \to T$ in distributions, then $\partial^\alpha T_n \to \partial^\alpha T$.*

Proof. Evaluate on a test function and move the derivative to that fixed test function; convergence of T_n then applies directly. ☐

16.10 Worked examples

Example 16.4 (Distributions diagnostic)**.** Give a rigorous worked analysis of Distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A distribution on Ω is a continuous linear functional on $C_c^\infty(\Omega)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 16.5 (Regular distributions diagnostic)**.** Give a rigorous worked analysis of Regular distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Locally integrable functions define distributions by integration against test functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 16.6 (Dirac mass diagnostic)**.** Give a rigorous worked analysis of Dirac mass in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Point evaluation $\delta_{x_0}(\phi) = \phi(x_0)$ is a distribution not represented by an ordinary locally integrable function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 16.7 (Distributional derivatives diagnostic)**.** Give a rigorous worked analysis of Distributional derivatives in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Define $\partial^\alpha T(\phi) = (-1)^{|\alpha|} T(\partial^\alpha \phi)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

16.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 16.8 (Distributions diagnostic)**.** Give a rigorous worked analysis of Distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A distribution on Ω is a continuous linear functional on $C_c^\infty(\Omega)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 16.9 (Regular distributions diagnostic)**.** Give a rigorous worked analysis of Regular distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Locally integrable functions define distributions by integration against test functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 16.10 (Dirac mass diagnostic)**.** Give a rigorous worked analysis of Dirac mass in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Point evaluation $\delta_{x_0}(\phi) = \phi(x_0)$ is a distribution not represented by an ordinary locally integrable function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 16.11 (Distributional derivatives diagnostic)**.** Give a rigorous worked analysis of Distributional derivatives in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Define $\partial^\alpha T(\phi) = (-1)^{|\alpha|} T(\partial^\alpha \phi)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 16.12 (Integration by parts diagnostic)**.** Give a rigorous worked analysis of Integration by parts in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For smooth functions, distributional differentiation agrees with classical differentiation because test functions have compact support. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 16.13 (Convergence of distributions diagnostic)**.** Give a rigorous worked analysis of Convergence of distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A sequence converges distributionally when it converges on every test function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 16.14 (Multiplication by smooth functions diagnostic)**.** Give a rigorous worked analysis of Multiplication by smooth functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. If $a \in C^\infty$, define $(aT)(\phi) = T(a\phi)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 16.15 (Linear differential operators diagnostic)**.** Give a rigorous worked analysis of Linear differential operators in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Smooth-coefficient differential operators act naturally on distributions by combining multiplication and distributional differentiation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 16.16 (Agreement with classical derivatives proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in C^1$ defines a regular distribution, then its distributional derivative is the regular distribution defined by f'.

Solution. For every test function, $D_f'(\phi) = -\int f\phi' = \int f'\phi$ by integration by parts and vanishing boundary terms. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 16.17 (Derivative of Heaviside proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: The distributional derivative of the Heaviside step is δ_0.

Solution. For a test function ϕ, $H'(\phi) = -\int_0^\infty \phi'(x)dx = \phi(0)$. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 16.18 (Continuity of differentiation proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $T_n \to T$ in distributions, then $\partial^\alpha T_n \to \partial^\alpha T$.

Solution. Evaluate on a test function and move the derivative to that fixed test function; convergence of T_n then applies directly. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 16.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Distributions extend the derivative and transform calculus to nonsmooth objects by duality. A distribution acts continuously and linearly on test functions, and derivatives are transferred onto the test function by integration by parts. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

16.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 16.20. State and apply the central principle behind Distributions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A distribution on Ω is a continuous linear functional on $C_c^\infty(\Omega)$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 16.21. State and apply the central principle behind Regular distributions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Locally integrable functions define distributions by integration against test functions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 16.22. State and apply the central principle behind Dirac mass. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Point evaluation $\delta_{x_0}(\phi) = \phi(x_0)$ is a distribution not represented by an ordinary locally integrable function. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 16.23. State and apply the central principle behind Distributional derivatives. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Define $\partial^\alpha T(\phi) = (-1)^{|\alpha|} T(\partial^\alpha \phi)$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 16.24. State and apply the central principle behind Integration by parts. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. For smooth functions, distributional differentiation agrees with classical differentiation because test functions have compact support. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 16.25. State and apply the central principle behind Convergence of distributions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. A sequence converges distributionally when it converges on every test function. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 16.26. State and apply the central principle behind Multiplication by smooth functions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. If $a \in C^\infty$, define $(aT)(\phi) = T(a\phi)$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 16.27. State and apply the central principle behind Linear differential operators. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Smooth-coefficient differential operators act naturally on distributions by combining multiplication and distributional differentiation. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 17

Support and Singular Distributions

Support records where a distribution can actually detect test functions. Singular distributions concentrate action on lower-dimensional or discrete sets and provide canonical models of sources, jumps, and point interactions.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

17.1 Support of a distribution

The support is the complement of the largest open set on which the distribution vanishes on every test function.

17.2 Compact support

Compactly supported distributions can act on broader classes of smooth functions by inserting a cutoff equal to one near the support.

17.3 Dirac support

The support of δ_{x_0} is the singleton $\{x_0\}$.

17.4 Derivatives of Dirac

$\delta^{(k)}$ acts by signed evaluation of the kth derivative of the test function.

17.5 Measures as distributions

Locally finite signed or complex measures define distributions by integration.

17.6 Jump singularities

Distributional derivatives of piecewise smooth functions include Dirac masses weighted by jump sizes.

17.7 Surface distributions

Integration over smooth hypersurfaces defines distributions supported on those hypersurfaces.

17.8 Localization

Multiplying by test functions or cutoffs isolates singular behavior near chosen points or sets.

17.9 Core structural results

Theorem 17.1 (Support of a derivative)**.** $\operatorname{supp}(\partial^\alpha T) \subseteq \operatorname{supp} T$.

Proof. If T vanishes on an open set, then it vanishes on derivatives of all test functions supported there, so the derivative distribution also vanishes there. ☐

Theorem 17.2 (Dirac derivatives)**.** $\delta_0^{(k)}(\phi) = (-1)^k \phi^{(k)}(0)$.

Proof. Apply the definition of distributional derivative repeatedly to the point-evaluation functional. ☐

Theorem 17.3 (Jump formula)**.** *If a piecewise C^1 function has jump J at x_0, its distributional derivative equals the classical derivative away from the jump plus $J\delta_{x_0}$.*

Proof. Integrate by parts separately on the two sides of the jump; the internal boundary terms combine to the jump size times $\phi(x_0)$. ☐

17.10 Worked examples

Example 17.4 (Support of a distribution diagnostic)**.** Give a rigorous worked analysis of Support of a distribution in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The support is the complement of the largest open set on which the distribution vanishes on every test function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 17.5 (Compact support diagnostic)**.** Give a rigorous worked analysis of Compact support in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Compactly supported distributions can act on broader classes of smooth functions by inserting a cutoff equal to one near the support. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 17.6 (Dirac support diagnostic)**.** Give a rigorous worked analysis of Dirac support in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The support of δ_{x_0} is the singleton $\{x_0\}$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 17.7 (Derivatives of Dirac diagnostic)**.** Give a rigorous worked analysis of Derivatives of Dirac in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. $\delta^{(k)}$ acts by signed evaluation of the kth derivative of the test function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

17.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 17.8 (Support of a distribution diagnostic)**.** Give a rigorous worked analysis of Support of a distribution in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The support is the complement of the largest open set on which the distribution vanishes on every test function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 17.9 (Compact support diagnostic)**.** Give a rigorous worked analysis of Compact support in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Compactly supported distributions can act on broader classes of smooth functions by inserting a cutoff equal to one near the support. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 17.10 (Dirac support diagnostic)**.** Give a rigorous worked analysis of Dirac support in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The support of δ_{x_0} is the singleton $\{x_0\}$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 17.11 (Derivatives of Dirac diagnostic)**.** Give a rigorous worked analysis of Derivatives of Dirac in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. $\delta^{(k)}$ acts by signed evaluation of the kth derivative of the test function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 17.12 (Measures as distributions diagnostic)**.** Give a rigorous worked analysis of Measures as distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Locally finite signed or complex measures define distributions by integration. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 17.13 (Jump singularities diagnostic)**.** Give a rigorous worked analysis of Jump singularities in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Distributional derivatives of piecewise smooth functions include Dirac masses weighted by jump sizes. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 17.14 (Surface distributions diagnostic)**.** Give a rigorous worked analysis of Surface distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Integration over smooth hypersurfaces defines distributions supported on those hypersurfaces. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 17.15 (Localization diagnostic)**.** Give a rigorous worked analysis of Localization in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Multiplying by test functions or cutoffs isolates singular behavior near chosen points or sets. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 17.16 (Support of a derivative proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: $\operatorname{supp}(\partial^\alpha T) \subseteq \operatorname{supp} T$.

Solution. If T vanishes on an open set, then it vanishes on derivatives of all test functions supported there, so the derivative distribution also vanishes there. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 17.17 (Dirac derivatives proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: $\delta_0^{(k)}(\phi) = (-1)^k \phi^{(k)}(0)$.

Solution. Apply the definition of distributional derivative repeatedly to the point-evaluation functional. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 17.18 (Jump formula proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If a piecewise C^1 function has jump J at x_0, its distributional derivative equals the classical derivative away from the jump plus $J\delta_{x_0}$.

Solution. Integrate by parts separately on the two sides of the jump; the internal boundary terms combine to the jump size times $\phi(x_0)$. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 17.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Support records where a distribution can actually detect test functions. Singular distributions concentrate action on lower-dimensional or discrete sets and provide canonical models of sources, jumps, and point interactions. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

17.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 17.20. State and apply the central principle behind Support of a distribution. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The support is the complement of the largest open set on which the distribution vanishes on every test function. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 17.21. State and apply the central principle behind Compact support. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Compactly supported distributions can act on broader classes of smooth functions by inserting a cutoff equal to one near the support. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 17.22. State and apply the central principle behind Dirac support. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The support of δ_{x_0} is the singleton $\{x_0\}$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 17.23. State and apply the central principle behind Derivatives of Dirac. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. $\delta^{(k)}$ acts by signed evaluation of the kth derivative of the test function. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 17.24. State and apply the central principle behind Measures as distributions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Locally finite signed or complex measures define distributions by integration. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 17.25. State and apply the central principle behind Jump singularities. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Distributional derivatives of piecewise smooth functions include Dirac masses weighted by jump sizes. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 17.26. State and apply the central principle behind Surface distributions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Integration over smooth hypersurfaces defines distributions supported on those hypersurfaces. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 17.27. State and apply the central principle behind Localization. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Multiplying by test functions or cutoffs isolates singular behavior near chosen points or sets. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 18

Tempered Distributions

Tempered distributions are continuous linear functionals on Schwartz space. They include polynomial-growth functions, measures, Dirac masses, and their derivatives while remaining stable under the Fourier transform.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

18.1 Tempered distributions

A tempered distribution is a continuous linear functional on $\mathcal{S}(\mathbb{R}^n)$.

18.2 Regular tempered distributions

Functions of at most polynomial growth define tempered distributions when paired with rapidly decaying Schwartz functions.

18.3 Schwartz topology

Continuity is measured against finitely many weighted derivative seminorms.

18.4 Differentiation

Distributional derivatives preserve temperedness.

18.5 Polynomial multiplication

Multiplication by coordinate polynomials preserves temperedness.

18.6 Dirac family

Dirac masses and all derivatives are tempered.

18.7 Measures

Many measures with polynomial growth define tempered distributions.

18.8 Fourier readiness

The Fourier invariance of Schwartz space lets the Fourier transform be transferred to its dual.

18.9 Core structural results

Theorem 18.1 (Polynomial-growth functions are tempered). *If $|f(x)| \leq C(1+|x|)^N$ and f is locally integrable, then f defines a tempered distribution.*

Proof. Choose a Schwartz seminorm with decay power larger than $N+n$; it dominates the integral of $|f\phi|$ and yields continuity. ☐

Theorem 18.2 (Derivatives preserve temperedness). *If $T \in \mathcal{S}'$, then $\partial^\alpha T \in \mathcal{S}'$.*

Proof. The map $\phi \mapsto \partial^\alpha \phi$ is continuous on Schwartz space, and composing it with the continuous functional T preserves continuity. ☐

Theorem 18.3 (Polynomial multiplication preserves temperedness). *If $T \in \mathcal{S}'$ and p is a polynomial, then $pT \in \mathcal{S}'$.*

Proof. Multiplication by a polynomial is a continuous endomorphism of Schwartz space, so $pT(\phi) = T(p\phi)$ is continuous. ☐

18.10 Worked examples

Example 18.4 (Tempered distributions diagnostic). Give a rigorous worked analysis of Tempered distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A tempered distribution is a continuous linear functional on $\mathcal{S}(\mathbb{R}^n)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 18.5 (Regular tempered distributions diagnostic). Give a rigorous worked analysis of Regular tempered distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Functions of at most polynomial growth define tempered distributions when paired with rapidly decaying Schwartz functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 18.6 (Schwartz topology diagnostic). Give a rigorous worked analysis of Schwartz topology in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Continuity is measured against finitely many weighted derivative seminorms. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 18.7 (Differentiation diagnostic)**.** Give a rigorous worked analysis of Differentiation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Distributional derivatives preserve temperedness. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

18.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 18.8 (Tempered distributions diagnostic)**.** Give a rigorous worked analysis of Tempered distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A tempered distribution is a continuous linear functional on $\mathcal{S}(\mathbb{R}^n)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 18.9 (Regular tempered distributions diagnostic)**.** Give a rigorous worked analysis of Regular tempered distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Functions of at most polynomial growth define tempered distributions when paired with rapidly decaying Schwartz functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 18.10 (Schwartz topology diagnostic)**.** Give a rigorous worked analysis of Schwartz topology in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Continuity is measured against finitely many weighted derivative seminorms. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 18.11 (Differentiation diagnostic)**.** Give a rigorous worked analysis of Differentiation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Distributional derivatives preserve temperedness. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 18.12 (Polynomial multiplication diagnostic)**.** Give a rigorous worked analysis of Polynomial multiplication in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Multiplication by coordinate polynomials preserves temperedness. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 18.13 (Dirac family diagnostic)**.** Give a rigorous worked analysis of Dirac family in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Dirac masses and all derivatives are tempered. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 18.14 (Measures diagnostic)**.** Give a rigorous worked analysis of Measures in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Many measures with polynomial growth define tempered distributions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 18.15 (Fourier readiness diagnostic)**.** Give a rigorous worked analysis of Fourier readiness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The Fourier invariance of Schwartz space lets the Fourier transform be transferred to its dual. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 18.16 (Polynomial-growth functions are tempered proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $|f(x)| \le C(1+|x|)^N$ and f is locally integrable, then f defines a tempered distribution.

Solution. Choose a Schwartz seminorm with decay power larger than $N+n$; it dominates the integral of $|f\phi|$ and yields continuity. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 18.17 (Derivatives preserve temperedness proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $T \in \mathcal{S}'$, then $\partial^\alpha T \in \mathcal{S}'$.

Solution. The map $\phi \mapsto \partial^\alpha \phi$ is continuous on Schwartz space, and composing it with the continuous functional T preserves continuity. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 18.18 (Polynomial multiplication preserves temperedness proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $T \in \mathcal{S}'$ and p is a polynomial, then $pT \in \mathcal{S}'$.

Solution. Multiplication by a polynomial is a continuous endomorphism of Schwartz space, so $pT(\phi) = T(p\phi)$ is continuous. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 18.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Tempered distributions are continuous linear functionals on Schwartz space. They include polynomial-growth functions, measures, Dirac masses, and their derivatives while remaining stable under the Fourier transform. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

18.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 18.20. State and apply the central principle behind Tempered distributions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. A tempered distribution is a continuous linear functional on $\mathcal{S}(\mathbb{R}^n)$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 18.21. State and apply the central principle behind Regular tempered distributions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Functions of at most polynomial growth define tempered distributions when paired with rapidly decaying Schwartz functions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 18.22. State and apply the central principle behind Schwartz topology. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Continuity is measured against finitely many weighted derivative seminorms. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 18.23. State and apply the central principle behind Differentiation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Distributional derivatives preserve temperedness. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 18.24. State and apply the central principle behind Polynomial multiplication. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Multiplication by coordinate polynomials preserves temperedness. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 18.25. State and apply the central principle behind Dirac family. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Dirac masses and all derivatives are tempered. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 18.26. State and apply the central principle behind Measures. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Many measures with polynomial growth define tempered distributions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 18.27. State and apply the central principle behind Fourier readiness. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The Fourier invariance of Schwartz space lets the Fourier transform be transferred to its dual. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 19

Fourier Transform of Distributions

The Fourier transform extends to tempered distributions by duality. The same algebraic rules for differentiation, multiplication, translation, and convolution then survive far beyond integrable functions.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

19.1 Dual definition

For $T \in \mathcal{S}'$, define $\widehat{T}(\phi) = T(\widehat{\phi})$ up to the convention-compatible inverse placement.

19.2 Fourier automorphism

Because the Fourier transform is a topological automorphism of Schwartz space, it induces an automorphism of tempered distributions.

19.3 Dirac transforms

Point masses transform into complex exponentials, while constants transform into Dirac masses under the standard convention.

19.4 Differentiation

Distributional differentiation becomes multiplication by polynomial frequency factors.

19.5 Polynomial multiplication

Multiplication by x^α becomes differentiation in frequency.

19.6 Translation and modulation

The classical transform identities extend by duality.

19.7 Convolution with Schwartz functions

A tempered distribution can be convolved with a Schwartz function to produce a smooth function with controlled growth.

19.8 PDE multipliers

Constant-coefficient differential equations become algebraic multiplier equations in frequency space.

19.9 Core structural results

Theorem 19.1 (Fourier transform of Dirac)**.** *Under the* 2π *convention,* $\widehat{\delta_0} = 1$ *as a tempered distribution.*

Proof. For a Schwartz test function, evaluate the dual definition and use Fourier inversion at zero to identify the resulting functional with integration against the constant one. ☐

Theorem 19.2 (Derivative rule)**.** $\widehat{\partial^\alpha T} = (2\pi i\xi)^\alpha \widehat{T}$.

Proof. Move the derivative onto the test function, use the classical Schwartz-space transform identity, and then return the multiplication to the transformed distribution. ☐

Theorem 19.3 (Fourier automorphism)**.** *The Fourier transform is a continuous bijection of* $\mathcal{S}'$ *with continuous inverse.*

Proof. Dualize the continuous bijection of Schwartz space and use the inverse Fourier transform for the inverse operator. ☐

19.10 Worked examples

Example 19.4 (Dual definition diagnostic)**.** Give a rigorous worked analysis of Dual definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For $T \in \mathcal{S}'$, define $\widehat{T}(\phi) = T(\widehat{\phi})$ up to the convention-compatible inverse placement. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 19.5 (Fourier automorphism diagnostic)**.** Give a rigorous worked analysis of Fourier automorphism in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Because the Fourier transform is a topological automorphism of Schwartz space, it induces an automorphism of tempered distributions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 19.6 (Dirac transforms diagnostic)**.** Give a rigorous worked analysis of Dirac transforms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Point masses transform into complex exponentials, while constants transform into Dirac masses under the standard convention. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 19.7 (Differentiation diagnostic)**.** Give a rigorous worked analysis of Differentiation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Distributional differentiation becomes multiplication by polynomial frequency factors. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

19.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 19.8 (Dual definition diagnostic)**.** Give a rigorous worked analysis of Dual definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For $T \in \mathcal{S}'$, define $\widehat{T}(\phi) = T(\widehat{\phi})$ up to the convention-compatible inverse placement. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 19.9 (Fourier automorphism diagnostic)**.** Give a rigorous worked analysis of Fourier automorphism in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Because the Fourier transform is a topological automorphism of Schwartz space, it induces an automorphism of tempered distributions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 19.10 (Dirac transforms diagnostic)**.** Give a rigorous worked analysis of Dirac transforms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Point masses transform into complex exponentials, while constants transform into Dirac masses under the standard convention. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 19.11 (Differentiation diagnostic)**.** Give a rigorous worked analysis of Differentiation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Distributional differentiation becomes multiplication by polynomial frequency factors. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 19.12 (Polynomial multiplication diagnostic)**.** Give a rigorous worked analysis of Polynomial multiplication in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Multiplication by x^α becomes differentiation in frequency. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 19.13 (Translation and modulation diagnostic)**.** Give a rigorous worked analysis of Translation and modulation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The classical transform identities extend by duality. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 19.14 (Convolution with Schwartz functions diagnostic)**.** Give a rigorous worked analysis of Convolution with Schwartz functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A tempered distribution can be convolved with a Schwartz function to produce a smooth function with controlled growth. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 19.15 (PDE multipliers diagnostic)**.** Give a rigorous worked analysis of PDE multipliers in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Constant-coefficient differential equations become algebraic multiplier equations in frequency space. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 19.16 (Fourier transform of Dirac proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: Under the 2π convention, $\widehat{\delta_0} = 1$ as a tempered distribution.

Solution. For a Schwartz test function, evaluate the dual definition and use Fourier inversion at zero to identify the resulting functional with integration against the constant one. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 19.17 (Derivative rule proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: $\widehat{\partial^\alpha T} = (2\pi i\xi)^\alpha \widehat{T}$.

Solution. Move the derivative onto the test function, use the classical Schwartz-space transform identity, and then return the multiplication to the transformed distribution. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 19.18 (Fourier automorphism proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: The Fourier transform is a continuous bijection of $\mathcal{S}'$ with continuous inverse.

Solution. Dualize the continuous bijection of Schwartz space and use the inverse Fourier transform for the inverse operator. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 19.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. The Fourier transform extends to tempered distributions by duality. The same algebraic rules for differentiation, multiplication, translation, and convolution then survive far beyond integrable functions. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. ☐

19.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 19.20. State and apply the central principle behind Dual definition. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. For $T \in \mathcal{S}'$, define $\widehat{T}(\phi) = T(\widehat{\phi})$ up to the convention-compatible inverse placement. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 19.21. State and apply the central principle behind Fourier automorphism. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Because the Fourier transform is a topological automorphism of Schwartz space, it induces an automorphism of tempered distributions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 19.22. State and apply the central principle behind Dirac transforms. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Point masses transform into complex exponentials, while constants transform into Dirac masses under the standard convention. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 19.23. State and apply the central principle behind Differentiation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Distributional differentiation becomes multiplication by polynomial frequency factors. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 19.24. State and apply the central principle behind Polynomial multiplication. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Multiplication by x^α becomes differentiation in frequency. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 19.25. State and apply the central principle behind Translation and modulation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The classical transform identities extend by duality. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 19.26. State and apply the central principle behind Convolution with Schwartz functions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A tempered distribution can be convolved with a Schwartz function to produce a smooth function with controlled growth. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 19.27. State and apply the central principle behind PDE multipliers. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Constant-coefficient differential equations become algebraic multiplier equations in frequency space. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Part IV

Sobolev and PDE Methods

Chapter 20

Weak Derivatives

Weak derivatives extend differentiation to L^1_{loc} functions by testing against compactly supported smooth functions. They are the bridge from distribution theory to Sobolev spaces and weak PDE formulations.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

20.1 Weak derivative definition

A function g is the weak derivative $D_j f$ if $\int f D_j \phi = -\int g\phi$ for every test function.

20.2 Agreement with classical derivatives

Classically differentiable functions have the same weak derivative.

20.3 Piecewise smooth functions

Jumps usually prevent the weak derivative from being represented by an ordinary function because the distributional derivative contains Dirac masses.

20.4 Uniqueness

Weak derivatives are unique up to almost-everywhere equality.

20.5 Higher weak derivatives

Iterating the definition yields multi-index weak derivatives.

20.6 Locality

Weak derivatives depend only on local almost-everywhere behavior.

20.7 Integration by parts

The weak derivative definition is precisely the boundary-free integration-by-parts identity.

20.8 Approximation

Mollification converts weak derivatives into classical derivatives and commutes with differentiation on interior regions.

20.9 Core structural results

Theorem 20.1 (Uniqueness of weak derivative)**.** *If $g, h \in L^1_{\mathrm{loc}}$ both satisfy the weak derivative identity for f, then $g = h$ almost everywhere.*

Proof. Their difference integrates to zero against every test function; the fundamental lemma of distributions implies the associated regular distribution is zero, hence the functions agree a.e. □

Theorem 20.2 (Classical implies weak)**.** *If $f \in C^1$, then its weak derivative equals its classical derivative.*

Proof. Integrate by parts against compactly supported test functions; boundary terms vanish. □

Theorem 20.3 (Mollifier commutation)**.** *For locally integrable f with weak derivative $D_j f$, $D_j(f * \rho_\varepsilon) = (D_j f) * \rho_\varepsilon$ where defined.*

Proof. Differentiate the smooth convolution and transfer the derivative from the mollifier to f using the weak derivative identity. □

20.10 Worked examples

Example 20.4 (Weak derivative definition diagnostic)**.** Give a rigorous worked analysis of Weak derivative definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A function g is the weak derivative $D_j f$ if $\int f D_j \phi = -\int g\phi$ for every test function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 20.5 (Agreement with classical derivatives diagnostic)**.** Give a rigorous worked analysis of Agreement with classical derivatives in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Classically differentiable functions have the same weak derivative. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 20.6 (Piecewise smooth functions diagnostic)**.** Give a rigorous worked analysis of Piecewise smooth functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Jumps usually prevent the weak derivative from being represented by an ordinary function because the distributional derivative contains Dirac masses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 20.7 (Uniqueness diagnostic)**.** Give a rigorous worked analysis of Uniqueness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Weak derivatives are unique up to almost-everywhere equality. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

20.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 20.8 (Weak derivative definition diagnostic)**.** Give a rigorous worked analysis of Weak derivative definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A function g is the weak derivative $D_j f$ if $\int f D_j \phi = -\int g\phi$ for every test function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 20.9 (Agreement with classical derivatives diagnostic)**.** Give a rigorous worked analysis of Agreement with classical derivatives in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Classically differentiable functions have the same weak derivative. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 20.10 (Piecewise smooth functions diagnostic)**.** Give a rigorous worked analysis of Piecewise smooth functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Jumps usually prevent the weak derivative from being represented by an ordinary function because the distributional derivative contains Dirac masses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 20.11 (Uniqueness diagnostic)**.** Give a rigorous worked analysis of Uniqueness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Weak derivatives are unique up to almost-everywhere equality. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 20.12 (Higher weak derivatives diagnostic)**.** Give a rigorous worked analysis of Higher weak derivatives in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Iterating the definition yields multi-index weak derivatives. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 20.13 (Locality diagnostic)**.** Give a rigorous worked analysis of Locality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Weak derivatives depend only on local almost-everywhere behavior. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 20.14 (Integration by parts diagnostic)**.** Give a rigorous worked analysis of Integration by parts in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The weak derivative definition is precisely the boundary-free integration-by-parts identity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 20.15 (Approximation diagnostic)**.** Give a rigorous worked analysis of Approximation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Mollification converts weak derivatives into classical derivatives and commutes with differentiation on interior regions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 20.16 (Uniqueness of weak derivative proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $g, h \in L^1_{\mathrm{loc}}$ both satisfy the weak derivative identity for f, then $g = h$ almost everywhere.

Solution. Their difference integrates to zero against every test function; the fundamental lemma of distributions implies the associated regular distribution is zero, hence the functions agree a.e. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 20.17 (Classical implies weak proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in C^1$, then its weak derivative equals its classical derivative.

Solution. Integrate by parts against compactly supported test functions; boundary terms vanish. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 20.18 (Mollifier commutation proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For locally integrable f with weak derivative $D_j f$, $D_j(f * \rho_\varepsilon) = (D_j f) * \rho_\varepsilon$ where defined.

Solution. Differentiate the smooth convolution and transfer the derivative from the mollifier to f using the weak derivative identity. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 20.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Weak derivatives extend differentiation to L^1_{loc} functions by testing against compactly supported smooth functions. They are the bridge from distribution theory to Sobolev spaces and weak PDE formulations. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. ☐

20.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 20.20. State and apply the central principle behind Weak derivative definition. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. A function g is the weak derivative $D_j f$ if $\int f D_j\phi = -\int g\phi$ for every test function. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 20.21. State and apply the central principle behind Agreement with classical derivatives. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Classically differentiable functions have the same weak derivative. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 20.22. State and apply the central principle behind Piecewise smooth functions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Jumps usually prevent the weak derivative from being represented by an ordinary function because the distributional derivative contains Dirac masses. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 20.23. State and apply the central principle behind Uniqueness. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Weak derivatives are unique up to almost-everywhere equality. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 20.24. State and apply the central principle behind Higher weak derivatives. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Iterating the definition yields multi-index weak derivatives. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 20.25. State and apply the central principle behind Locality. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Weak derivatives depend only on local almost-everywhere behavior. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 20.26. State and apply the central principle behind Integration by parts. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. The weak derivative definition is precisely the boundary-free integration-by-parts identity. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 20.27. State and apply the central principle behind Approximation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Mollification converts weak derivatives into classical derivatives and commutes with differentiation on interior regions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 21

Sobolev Spaces

Sobolev spaces measure both a function and its weak derivatives in L^p. They provide the natural energy spaces for variational PDEs, compactness arguments, and weak boundary conditions.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

21.1 Definition of $W^{k,p}$

A function lies in $W^{k,p}(\Omega)$ when all weak derivatives through order k belong to L^p.

21.2 Sobolev norms

Sum the L^p norms of derivatives up to order k; for $p = 2$ this gives a Hilbert-space norm.

21.3 H^k notation

$H^k = W^{k,2}$ is especially important because inner-product methods apply.

21.4 Completeness

Sobolev spaces are Banach spaces, and H^k spaces are Hilbert spaces.

21.5 Zero-boundary spaces

$W_0^{1,p}$ is the closure of compactly supported smooth functions and encodes homogeneous Dirichlet boundary data.

21.6 Poincare inequality

On suitable bounded domains, the L^p norm of a zero-boundary function is controlled by its gradient.

21.7 Embeddings

Sufficient Sobolev regularity implies improved integrability or continuity.

21.8 Compactness

Rellich-type theorems turn bounded Sobolev sequences into strongly convergent subsequences in weaker norms.

21.9 Core structural results

Theorem 21.1 (Completeness of $W^{k,p}$)**.** *For $1 \leq p \leq \infty$, $W^{k,p}$ is complete.*

Proof. A Cauchy sequence is Cauchy in L^p for every derivative component; pass to limits in each component and use distributional convergence to verify that the limiting components are the weak derivatives of the limiting function. □

Theorem 21.2 (Poincare inequality)**.** *On a bounded Lipschitz domain, $\|u\|_{L^p} \leq C\|\nabla u\|_{L^p}$ for $u \in W_0^{1,p}$.*

Proof. Prove first for smooth compactly supported functions by integrating along suitable line segments or using contradiction and compactness, then extend by density. □

Theorem 21.3 (Hilbert structure of H^k)**.** *H^k is a Hilbert space with inner product obtained by summing L^2 inner products of weak derivatives through order k.*

Proof. Completeness follows from Sobolev completeness and the norm is induced by the stated positive-definite inner product. □

21.10 Worked examples

Example 21.4 (Definition of $W^{k,p}$ diagnostic)**.** Give a rigorous worked analysis of Definition of $W^{k,p}$ in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A function lies in $W^{k,p}(\Omega)$ when all weak derivatives through order k belong to L^p. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 21.5 (Sobolev norms diagnostic)**.** Give a rigorous worked analysis of Sobolev norms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Sum the L^p norms of derivatives up to order k; for $p = 2$ this gives a Hilbert-space norm. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 21.6 (H^k notation diagnostic)**.** Give a rigorous worked analysis of H^k notation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. $H^k = W^{k,2}$ is especially important because inner-product methods apply. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 21.7 (Completeness diagnostic)**.** Give a rigorous worked analysis of Completeness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Sobolev spaces are Banach spaces, and H^k spaces are Hilbert spaces. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

21.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 21.8 (Definition of $W^{k,p}$ diagnostic)**.** Give a rigorous worked analysis of Definition of $W^{k,p}$ in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A function lies in $W^{k,p}(\Omega)$ when all weak derivatives through order k belong to L^p. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 21.9 (Sobolev norms diagnostic)**.** Give a rigorous worked analysis of Sobolev norms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Sum the L^p norms of derivatives up to order k; for $p = 2$ this gives a Hilbert-space norm. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 21.10 (H^k notation diagnostic)**.** Give a rigorous worked analysis of H^k notation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. $H^k = W^{k,2}$ is especially important because inner-product methods apply. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 21.11 (Completeness diagnostic)**.** Give a rigorous worked analysis of Completeness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Sobolev spaces are Banach spaces, and H^k spaces are Hilbert spaces. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 21.12 (Zero-boundary spaces diagnostic)**.** Give a rigorous worked analysis of Zero-boundary spaces in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. $W_0^{1,p}$ is the closure of compactly supported smooth functions and encodes homogeneous Dirichlet boundary data. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 21.13 (Poincare inequality diagnostic)**.** Give a rigorous worked analysis of Poincare inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. On suitable bounded domains, the L^p norm of a zero-boundary function is controlled by its gradient. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 21.14 (Embeddings diagnostic)**.** Give a rigorous worked analysis of Embeddings in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Sufficient Sobolev regularity implies improved integrability or continuity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 21.15 (Compactness diagnostic)**.** Give a rigorous worked analysis of Compactness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Rellich-type theorems turn bounded Sobolev sequences into strongly convergent subsequences in weaker norms. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 21.16 (Completeness of $W^{k,p}$ proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For $1 \le p \le \infty$, $W^{k,p}$ is complete.

Solution. A Cauchy sequence is Cauchy in L^p for every derivative component; pass to limits in each component and use distributional convergence to verify that the limiting components are the weak derivatives of the limiting function. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 21.17 (Poincare inequality proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: On a bounded Lipschitz domain, $\|u\|_{L^p} \le C\|\nabla u\|_{L^p}$ for $u \in W_0^{1,p}$.

Solution. Prove first for smooth compactly supported functions by integrating along suitable line segments or using contradiction and compactness, then extend by density. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 21.18 (Hilbert structure of H^k proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: H^k is a Hilbert space with inner product obtained by summing L^2 inner products of weak derivatives through order k.

Solution. Completeness follows from Sobolev completeness and the norm is induced by the stated positive-definite inner product. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 21.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Sobolev spaces measure both a function and its weak derivatives in L^p. They provide the natural energy spaces for variational PDEs, compactness arguments, and weak boundary conditions. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

21.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 21.20. State and apply the central principle behind Definition of $W^{k,p}$. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. $\square$

Solution. A function lies in $W^{k,p}(\Omega)$ when all weak derivatives through order k belong to L^p. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 21.21. State and apply the central principle behind Sobolev norms. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. $\square$

Solution. Sum the L^p norms of derivatives up to order k; for $p = 2$ this gives a Hilbert-space norm. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 21.22. State and apply the central principle behind H^k notation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. $\square$

Solution. $H^k = W^{k,2}$ is especially important because inner-product methods apply. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 21.23. State and apply the central principle behind Completeness. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. $\square$

Solution. Sobolev spaces are Banach spaces, and H^k spaces are Hilbert spaces. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 21.24. State and apply the central principle behind Zero-boundary spaces. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. $\square$

Solution. $W_0^{1,p}$ is the closure of compactly supported smooth functions and encodes homogeneous Dirichlet boundary data. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 21.25. State and apply the central principle behind Poincare inequality. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. On suitable bounded domains, the L^p norm of a zero-boundary function is controlled by its gradient. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 21.26. State and apply the central principle behind Embeddings. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Sufficient Sobolev regularity implies improved integrability or continuity. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 21.27. State and apply the central principle behind Compactness. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Rellich-type theorems turn bounded Sobolev sequences into strongly convergent subsequences in weaker norms. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 22

Approximation and Density

Approximation and density justify replacing rough functions by smooth ones. Mollifiers, cutoffs, and extension arguments let proofs and numerical constructions move between Sobolev or L^p functions and smooth compactly supported models.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

22.1 Mollifiers

A standard mollifier is smooth, nonnegative, compactly supported, and normalized to integral one.

22.2 Approximate identities

Scaled mollifiers concentrate at the origin and approximate functions in L^p.

22.3 Smooth approximation in L^p

Convolution plus cutoff proves density of C_c^∞ in L^p for $1 \le p < \infty$.

22.4 Sobolev approximation

Mollification approximates weak derivatives simultaneously on interior subdomains.

22.5 Boundary issues

Near the boundary, extension operators or geometry-aware cutoffs are needed to mollify without leaving the domain.

22.6 Density of test functions

$C_c^\infty(\Omega)$ is dense in $W_0^{1,p}(\Omega)$ by definition and under standard characterizations.

22.7 Frequency truncation

Fourier cutoffs give an alternative smooth approximation method in whole-space Sobolev spaces.

22.8 Approximation consequences

Density transfers identities and inequalities proved for smooth functions to rough functions by norm limits.

22.9 Core structural results

Theorem 22.1 (L^p mollifier convergence)**.** *If $f \in L^p(\mathbb{R}^n)$, $1 \le p < \infty$, then $f * \rho_\varepsilon \to f$ in L^p.*

Proof. Write the convolution difference as an average of translations $\tau_y f - f$ and use continuity of translations in L^p together with concentration of the mollifier near zero. ☐

Theorem 22.2 (Derivative commutation)**.** *If $f \in W^{k,p}$, then $D^\alpha(f * \rho_\varepsilon) = (D^\alpha f) * \rho_\varepsilon$ for $|\alpha| \le k$.*

Proof. Use the weak derivative identity to move derivatives between the function and mollifier, then apply convolution associativity on compactly supported kernels. ☐

Theorem 22.3 (Density of C_c^∞ in L^p)**.** *For $1 \le p < \infty$, smooth compactly supported functions are dense in $L^p(\mathbb{R}^n)$.*

Proof. First truncate f spatially and in magnitude to obtain bounded finite-support approximants, then mollify them; each step has arbitrarily small L^p error. ☐

22.10 Worked examples

Example 22.4 (Mollifiers diagnostic)**.** Give a rigorous worked analysis of Mollifiers in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A standard mollifier is smooth, nonnegative, compactly supported, and normalized to integral one. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 22.5 (Approximate identities diagnostic)**.** Give a rigorous worked analysis of Approximate identities in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Scaled mollifiers concentrate at the origin and approximate functions in L^p. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 22.6 (Smooth approximation in L^p diagnostic)**.** Give a rigorous worked analysis of Smooth approximation in L^p in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Convolution plus cutoff proves density of C_c^∞ in

L^p for $1 \leq p < \infty$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 22.7 (Sobolev approximation diagnostic)**.** Give a rigorous worked analysis of Sobolev approximation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Mollification approximates weak derivatives simultaneously on interior subdomains. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

22.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 22.8 (Mollifiers diagnostic)**.** Give a rigorous worked analysis of Mollifiers in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A standard mollifier is smooth, nonnegative, compactly supported, and normalized to integral one. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 22.9 (Approximate identities diagnostic)**.** Give a rigorous worked analysis of Approximate identities in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Scaled mollifiers concentrate at the origin and approximate functions in L^p. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 22.10 (Smooth approximation in L^p diagnostic)**.** Give a rigorous worked analysis of Smooth approximation in L^p in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Convolution plus cutoff proves density of C_c^∞ in L^p for $1 \leq p < \infty$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 22.11 (Sobolev approximation diagnostic)**.** Give a rigorous worked analysis of Sobolev approximation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Mollification approximates weak derivatives simultaneously on interior subdomains. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 22.12 (Boundary issues diagnostic)**.** Give a rigorous worked analysis of Boundary issues in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Near the boundary, extension operators or geometry-aware cutoffs are needed to mollify without leaving the domain. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 22.13 (Density of test functions diagnostic)**.** Give a rigorous worked analysis of Density of test functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. $C_c^\infty(\Omega)$ is dense in $W_0^{1,p}(\Omega)$ by definition and under standard characterizations. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 22.14 (Frequency truncation diagnostic)**.** Give a rigorous worked analysis of Frequency truncation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Fourier cutoffs give an alternative smooth approximation method in whole-space Sobolev spaces. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 22.15 (Approximation consequences diagnostic)**.** Give a rigorous worked analysis of Approximation consequences in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Density transfers identities and inequalities proved for smooth functions to rough functions by norm limits. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 22.16 (L^p mollifier convergence proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in L^p(\mathbb{R}^n)$, $1 \le p < \infty$, then $f * \rho_\varepsilon \to f$ in L^p.

Solution. Write the convolution difference as an average of translations $\tau_y f - f$ and use continuity of translations in L^p together with concentration of the mollifier near zero. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 22.17 (Derivative commutation proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in W^{k,p}$, then $D^\alpha(f * \rho_\varepsilon) = (D^\alpha f) * \rho_\varepsilon$ for $|\alpha| \le k$.

Solution. Use the weak derivative identity to move derivatives between the function and mollifier, then apply convolution associativity on compactly supported kernels. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 22.18 (Density of C_c^∞ in L^p proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For $1 \le p < \infty$, smooth compactly supported functions are dense in $L^p(\mathbb{R}^n)$.

Solution. First truncate f spatially and in magnitude to obtain bounded finite-support approximants, then mollify them; each step has arbitrarily small L^p error. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 22.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Approximation and density justify replacing rough functions by smooth ones. Mollifiers, cutoffs, and extension arguments let proofs and numerical constructions move between Sobolev or L^p functions and smooth compactly supported models. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. ☐

22.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 22.20. State and apply the central principle behind Mollifiers. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. A standard mollifier is smooth, nonnegative, compactly supported, and normalized to integral one. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 22.21. State and apply the central principle behind Approximate identities. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Scaled mollifiers concentrate at the origin and approximate functions in L^p. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 22.22. State and apply the central principle behind Smooth approximation in L^p. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Convolution plus cutoff proves density of C_c^∞ in L^p for $1 \leq p < \infty$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 22.23. State and apply the central principle behind Sobolev approximation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Mollification approximates weak derivatives simultaneously on interior subdomains. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 22.24. State and apply the central principle behind Boundary issues. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Near the boundary, extension operators or geometry-aware cutoffs are needed to mollify without leaving the domain. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 22.25. State and apply the central principle behind Density of test functions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. $C_c^\infty(\Omega)$ is dense in $W_0^{1,p}(\Omega)$ by definition and under standard characterizations. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 22.26. State and apply the central principle behind Frequency truncation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Fourier cutoffs give an alternative smooth approximation method in whole-space Sobolev spaces. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 22.27. State and apply the central principle behind Approximation consequences. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Density transfers identities and inequalities proved for smooth functions to rough functions by norm limits. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 23

Weak Boundary-Value Problems

Weak boundary-value problems replace pointwise differential equations by integral identities tested against admissible functions. This is the natural formulation for rough coefficients, Sobolev solutions, and variational existence theory.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

23.1 Strong and weak formulations

A strong solution satisfies the PDE pointwise; a weak solution satisfies an integrated identity obtained by multiplying by test functions and integrating by parts.

23.2 Energy spaces

Sobolev spaces such as $H_0^1(\Omega)$ encode square-integrable weak gradients and homogeneous boundary conditions.

23.3 Bilinear forms

Second-order elliptic equations are represented by bilinear forms such as $a(u, v) = \int_\Omega A\nabla u \cdot \nabla v + cuv$.

23.4 Linear functionals

Forcing terms act as continuous linear functionals on the chosen energy space.

23.5 Coercivity

A coercive estimate $a(v, v) \geq c\|v\|_V^2$ supplies control of the solution norm.

23.6 Weak boundary conditions

Dirichlet conditions are built into H_0^1; Neumann data typically arise through boundary terms in integration by parts.

23.7 Galerkin approximation

Finite-dimensional subspaces convert the weak problem into linear algebra and support constructive approximation.

23.8 Uniqueness by energy

Testing the difference of two solutions against itself often proves uniqueness from positivity or coercivity.

23.9 Core structural results

Theorem 23.1 (Lax–Milgram theorem)**.** *If V is a Hilbert space, a is bounded and coercive, and $F \in V^*$, then there is a unique $u \in V$ with $a(u, v) = F(v)$ for all $v \in V$.*

Proof. Use Riesz representation to write $a(u, v) = \langle Au, v\rangle$ for a bounded operator A. Coercivity makes A bounded below, hence injective with closed range; orthogonality to the range would contradict coercivity for the adjoint, so the range is all of V. □

Theorem 23.2 (Energy uniqueness)**.** *If a is coercive and $a(u, v) = 0$ for all v, then $u = 0$.*

Proof. Choose $v = u$. Then $0 = a(u, u) \geq c\|u\|_V^2$, hence $u = 0$. □

Theorem 23.3 (Galerkin stability)**.** *For coercive a, the Galerkin solution u_h satisfies $\|u_h\|_V \leq c^{-1}\|F\|_{V^*}$.*

Proof. Test the discrete equation with u_h and combine coercivity with the dual norm bound. □

23.10 Worked examples

Example 23.4 (Strong and weak formulations diagnostic)**.** Give a rigorous worked analysis of Strong and weak formulations in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A strong solution satisfies the PDE pointwise; a weak solution satisfies an integrated identity obtained by multiplying by test functions and integrating by parts. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 23.5 (Energy spaces diagnostic)**.** Give a rigorous worked analysis of Energy spaces in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Sobolev spaces such as $H_0^1(\Omega)$ encode square-integrable weak gradients and homogeneous boundary conditions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 23.6 (Bilinear forms diagnostic)**.** Give a rigorous worked analysis of Bilinear forms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Second-order elliptic equations are represented by bilinear forms such as $a(u, v) = \int_\Omega A\nabla u \cdot \nabla v + cuv$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 23.7 (Linear functionals diagnostic)**.** Give a rigorous worked analysis of Linear functionals in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Forcing terms act as continuous linear functionals on the chosen energy space. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

23.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 23.8 (Strong and weak formulations diagnostic)**.** Give a rigorous worked analysis of Strong and weak formulations in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A strong solution satisfies the PDE pointwise; a weak solution satisfies an integrated identity obtained by multiplying by test functions and integrating by parts. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 23.9 (Energy spaces diagnostic)**.** Give a rigorous worked analysis of Energy spaces in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Sobolev spaces such as $H_0^1(\Omega)$ encode square-integrable weak gradients and homogeneous boundary conditions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 23.10 (Bilinear forms diagnostic)**.** Give a rigorous worked analysis of Bilinear forms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Second-order elliptic equations are represented by bilinear forms such as $a(u, v) = \int_\Omega A\nabla u \cdot \nabla v + cuv$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 23.11 (Linear functionals diagnostic)**.** Give a rigorous worked analysis of Linear functionals in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Forcing terms act as continuous linear functionals on the chosen energy space. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 23.12 (Coercivity diagnostic)**.** Give a rigorous worked analysis of Coercivity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A coercive estimate $a(v, v) \geq c\|v\|_V^2$ supplies control of the solution norm. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 23.13 (Weak boundary conditions diagnostic)**.** Give a rigorous worked analysis of Weak boundary conditions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Dirichlet conditions are built into H_0^1; Neumann data typically arise through boundary terms in integration by parts. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 23.14 (Galerkin approximation diagnostic)**.** Give a rigorous worked analysis of Galerkin approximation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Finite-dimensional subspaces convert the weak problem into linear algebra and support constructive approximation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 23.15 (Uniqueness by energy diagnostic)**.** Give a rigorous worked analysis of Uniqueness by energy in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Testing the difference of two solutions against itself often proves uniqueness from positivity or coercivity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 23.16 (Lax–Milgram theorem proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If V is a Hilbert space, a is bounded and coercive, and $F \in V^*$, then there is a unique $u \in V$ with $a(u, v) = F(v)$ for all $v \in V$.

Solution. Use Riesz representation to write $a(u, v) = \langle Au, v\rangle$ for a bounded operator A. Coercivity makes A bounded below, hence injective with closed range; orthogonality to the range would contradict coercivity for the adjoint, so the range is all of V. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 23.17 (Energy uniqueness proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If a is coercive and $a(u, v) = 0$ for all v, then $u = 0$.

Solution. Choose $v = u$. Then $0 = a(u, u) \geq c\|u\|_V^2$, hence $u = 0$. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 23.18 (Galerkin stability proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For coercive a, the Galerkin solution u_h satisfies $\|u_h\|_V \leq c^{-1}\|F\|_{V^*}$.

Solution. Test the discrete equation with u_h and combine coercivity with the dual norm bound. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 23.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Weak boundary-value problems replace pointwise differential equations by integral identities tested against admissible functions. This is the natural formulation for rough coefficients, Sobolev solutions, and variational existence theory. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

23.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 23.20. State and apply the central principle behind Strong and weak formulations. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A strong solution satisfies the PDE pointwise; a weak solution satisfies an integrated identity obtained by multiplying by test functions and integrating by parts. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 23.21. State and apply the central principle behind Energy spaces. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Sobolev spaces such as $H_0^1(\Omega)$ encode square-integrable weak gradients and homogeneous boundary conditions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 23.22. State and apply the central principle behind Bilinear forms. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Second-order elliptic equations are represented by bilinear forms such as $a(u, v) = \int_\Omega A\nabla u \cdot \nabla v + cuv$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 23.23. State and apply the central principle behind Linear functionals. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Forcing terms act as continuous linear functionals on the chosen energy space. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 23.24. State and apply the central principle behind Coercivity. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A coercive estimate $a(v, v) \geq c\|v\|_V^2$ supplies control of the solution norm. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 23.25. State and apply the central principle behind Weak boundary conditions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Dirichlet conditions are built into H_0^1; Neumann data typically arise through boundary terms in integration by parts. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 23.26. State and apply the central principle behind Galerkin approximation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Finite-dimensional subspaces convert the weak problem into linear algebra and support constructive approximation. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 23.27. State and apply the central principle behind Uniqueness by energy. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Testing the difference of two solutions against itself often proves uniqueness from positivity or coercivity. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 24

Fundamental Solutions

A fundamental solution is a distributional inverse kernel for a differential operator. Once one is known, convolution converts a constant-coefficient PDE into an explicit representation formula.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

24.1 Definition

A fundamental solution E for $P(D)$ satisfies $P(D)E = \delta_0$ in distributions.

24.2 Convolution solution

When convolution is legitimate, $u = E * f$ satisfies $P(D)u = f$.

24.3 Laplacian kernels

The Laplacian has logarithmic fundamental solution in two dimensions and Newtonian power kernels in dimensions at least three.

24.4 Heat kernel

The heat kernel is the spacetime propagation kernel for the heat operator with causal structure.

24.5 Wave kernels

Wave fundamental solutions reflect finite propagation and dimension-dependent support geometry.

24.6 Distributional verification

The singularity at the origin is detected by integrating by parts on punctured domains and passing to a boundary-flux limit.

24.7 Nonuniqueness

Adding any homogeneous solution to a fundamental solution yields another fundamental solution.

24.8 Regularity away from the pole

Fundamental solutions are smooth wherever the source point is absent.

24.9 Core structural results

Theorem 24.1 (Convolution inversion)**.** *If $P(D)E = \delta_0$ and the relevant convolutions exist, then $P(D)(E * f) = f$.*

Proof. Constant-coefficient differentiation commutes with convolution, so $P(D)(E * f) = (P(D)E) * f = \delta_0 * f = f$. □

Theorem 24.2 (Newtonian fundamental solution)**.** *For $n \geq 3$, a constant multiple of $|x|^{2-n}$ is a fundamental solution of $-\Delta$.*

Proof. The function is harmonic away from zero. Integrate against a test function on the complement of a small ball, integrate by parts, and choose the constant so the shrinking boundary flux tends to the test value at zero. □

Theorem 24.3 (Smoothness off the singularity)**.** *If $P(D)E = \delta_0$, then $P(D)E = 0$ on every open set avoiding the origin.*

Proof. Test functions supported away from zero annihilate δ_0, so the distribution solves the homogeneous equation there; standard regularity for the explicit classical kernels gives smoothness. □

24.10 Worked examples

Example 24.4 (Definition diagnostic)**.** Give a rigorous worked analysis of Definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A fundamental solution E for $P(D)$ satisfies $P(D)E = \delta_0$ in distributions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 24.5 (Convolution solution diagnostic)**.** Give a rigorous worked analysis of Convolution solution in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. When convolution is legitimate, $u = E * f$ satisfies $P(D)u = f$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 24.6 (Laplacian kernels diagnostic)**.** Give a rigorous worked analysis of Laplacian kernels in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The Laplacian has logarithmic fundamental solution in two dimensions and Newtonian power kernels in dimensions at least three. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 24.7 (Heat kernel diagnostic)**.** Give a rigorous worked analysis of Heat kernel in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The heat kernel is the spacetime propagation kernel for the heat operator with causal structure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

24.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 24.8 (Definition diagnostic)**.** Give a rigorous worked analysis of Definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A fundamental solution E for $P(D)$ satisfies $P(D)E = \delta_0$ in distributions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 24.9 (Convolution solution diagnostic)**.** Give a rigorous worked analysis of Convolution solution in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. When convolution is legitimate, $u = E * f$ satisfies $P(D)u = f$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 24.10 (Laplacian kernels diagnostic)**.** Give a rigorous worked analysis of Laplacian kernels in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The Laplacian has logarithmic fundamental solution in two dimensions and Newtonian power kernels in dimensions at least three. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 24.11 (Heat kernel diagnostic)**.** Give a rigorous worked analysis of Heat kernel in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The heat kernel is the spacetime propagation kernel for the heat operator with causal structure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 24.12 (Wave kernels diagnostic)**.** Give a rigorous worked analysis of Wave kernels in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Wave fundamental solutions reflect finite propagation and dimension-dependent support geometry. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 24.13 (Distributional verification diagnostic)**.** Give a rigorous worked analysis of Distributional verification in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The singularity at the origin is detected by integrating by parts on punctured domains and passing to a boundary-flux limit. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 24.14 (Nonuniqueness diagnostic)**.** Give a rigorous worked analysis of Nonuniqueness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Adding any homogeneous solution to a fundamental solution yields another fundamental solution. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 24.15 (Regularity away from the pole diagnostic)**.** Give a rigorous worked analysis of Regularity away from the pole in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Fundamental solutions are smooth wherever the source point is absent. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 24.16 (Convolution inversion proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $P(D)E = \delta_0$ and the relevant convolutions exist, then $P(D)(E * f) = f$.

Solution. Constant-coefficient differentiation commutes with convolution, so $P(D)(E * f) = (P(D)E) * f = \delta_0 * f = f$. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 24.17 (Newtonian fundamental solution proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For $n \geq 3$, a constant multiple of $|x|^{2-n}$ is a fundamental solution of $-\Delta$.

Solution. The function is harmonic away from zero. Integrate against a test function on the complement of a small ball, integrate by parts, and choose the constant so the shrinking boundary flux tends to the test value at zero. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 24.18 (Smoothness off the singularity proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $P(D)E = \delta_0$, then $P(D)E = 0$ on every open set avoiding the origin.

Solution. Test functions supported away from zero annihilate δ_0, so the distribution solves the homogeneous equation there; standard regularity for the explicit classical kernels gives smoothness. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 24.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. A fundamental solution is a distributional inverse kernel for a differential operator. Once one is known, convolution converts a constant-coefficient PDE into an explicit representation formula. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

24.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 24.20. State and apply the central principle behind Definition. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A fundamental solution E for $P(D)$ satisfies $P(D)E = \delta_0$ in distributions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 24.21. State and apply the central principle behind Convolution solution. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. When convolution is legitimate, $u = E * f$ satisfies $P(D)u = f$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 24.22. State and apply the central principle behind Laplacian kernels. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The Laplacian has logarithmic fundamental solution in two dimensions and Newtonian power kernels in dimensions at least three. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 24.23. State and apply the central principle behind Heat kernel. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The heat kernel is the spacetime propagation kernel for the heat operator with causal structure. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 24.24. State and apply the central principle behind Wave kernels. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Wave fundamental solutions reflect finite propagation and dimension-dependent support geometry. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 24.25. State and apply the central principle behind Distributional verification. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. The singularity at the origin is detected by integrating by parts on punctured domains and passing to a boundary-flux limit. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 24.26. State and apply the central principle behind Nonuniqueness. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Adding any homogeneous solution to a fundamental solution yields another fundamental solution. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 24.27. State and apply the central principle behind Regularity away from the pole. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Fundamental solutions are smooth wherever the source point is absent. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 25

Green Functions

Green functions adapt fundamental solutions to domains and boundary conditions. They separate the singular response to a point source from a correction chosen to enforce the boundary data.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

25.1 Green function definition

For a domain operator, $G(x, y)$ solves the differential equation in x with a point source at y and prescribed boundary behavior.

25.2 Singular plus regular decomposition

Typically $G(x, y) = E(x - y) - H(x, y)$, where E is a whole-space fundamental solution and H corrects the boundary values.

25.3 Representation formula

Solutions are expressed by integrating the forcing against G, plus boundary terms when nonhomogeneous data are present.

25.4 Symmetry

Self-adjoint elliptic problems often yield $G(x, y) = G(y, x)$.

25.5 Positivity

Maximum principles can imply nonnegativity of Dirichlet Green functions for positive elliptic operators.

25.6 One-dimensional construction

For second-order ODEs, Green kernels are assembled from two homogeneous solutions matched continuously with a derivative jump.

25.7 Boundary dependence

Changing Dirichlet to Neumann conditions changes the correction and may require compatibility conditions.

25.8 Operator inverse viewpoint

The Green kernel is the integral kernel of the inverse or resolvent of the boundary-value operator.

25.9 Core structural results

Theorem 25.1 (Green representation). *If G is a Dirichlet Green function for L and $Lu = f$, then formally $u(x) = \int_\Omega G(x,y)f(y)dy$ for homogeneous boundary data.*

Proof. Apply Green's identity to u and $G(x,\cdot)$, use the point-source equation for G, and use the boundary condition to eliminate the boundary term. □

Theorem 25.2 (Symmetry for self-adjoint problems). *Under suitable self-adjoint boundary conditions, $G(x,y) = G(y,x)$.*

Proof. Apply Green's identity to the two Green functions with poles at x and y, excise small neighborhoods of the poles, and pass to the limit; boundary terms vanish by the shared boundary condition. □

Theorem 25.3 (Derivative jump in one dimension). *For $-u'' = f$ on an interval, the Green kernel is continuous across $x = y$ and its first derivative has the jump required to create a Dirac mass.*

Proof. Integrating the equation $-G_{xx} = \delta_y$ across a small interval around y gives $-(G_x(y^+) - G_x(y^-)) = 1$, while absence of a delta-prime term forces continuity. □

25.10 Worked examples

Example 25.4 (Green function definition diagnostic). Give a rigorous worked analysis of Green function definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For a domain operator, $G(x,y)$ solves the differential equation in x with a point source at y and prescribed boundary behavior. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 25.5 (Singular plus regular decomposition diagnostic)**.** Give a rigorous worked analysis of Singular plus regular decomposition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Typically $G(x, y) = E(x - y) - H(x, y)$, where E is a whole-space fundamental solution and H corrects the boundary values. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 25.6 (Representation formula diagnostic)**.** Give a rigorous worked analysis of Representation formula in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Solutions are expressed by integrating the forcing against G, plus boundary terms when nonhomogeneous data are present. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 25.7 (Symmetry diagnostic)**.** Give a rigorous worked analysis of Symmetry in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Self-adjoint elliptic problems often yield $G(x, y) = G(y, x)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

25.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 25.8 (Green function definition diagnostic)**.** Give a rigorous worked analysis of Green function definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For a domain operator, $G(x, y)$ solves the differential equation in x with a point source at y and prescribed boundary behavior. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 25.9 (Singular plus regular decomposition diagnostic)**.** Give a rigorous worked analysis of Singular plus regular decomposition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Typically $G(x, y) = E(x - y) - H(x, y)$, where E is a whole-space fundamental solution and H corrects the boundary values. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 25.10 (Representation formula diagnostic)**.** Give a rigorous worked analysis of Representation formula in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Solutions are expressed by integrating the forcing against G, plus boundary terms when nonhomogeneous data are present. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 25.11 (Symmetry diagnostic)**.** Give a rigorous worked analysis of Symmetry in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Self-adjoint elliptic problems often yield $G(x, y) = G(y, x)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 25.12 (Positivity diagnostic)**.** Give a rigorous worked analysis of Positivity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Maximum principles can imply nonnegativity of Dirichlet Green functions for positive elliptic operators. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 25.13 (One-dimensional construction diagnostic)**.** Give a rigorous worked analysis of One-dimensional construction in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For second-order ODEs, Green kernels are assembled from two homogeneous solutions matched continuously with a derivative jump. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 25.14 (Boundary dependence diagnostic)**.** Give a rigorous worked analysis of Boundary dependence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Changing Dirichlet to Neumann conditions changes the correction and may require compatibility conditions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 25.15 (Operator inverse viewpoint diagnostic)**.** Give a rigorous worked analysis of Operator inverse viewpoint in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The Green kernel is the integral kernel of the inverse or resolvent of the boundary-value operator. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 25.16 (Green representation proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If G is a Dirichlet Green function for L and $Lu = f$, then formally $u(x) = \int_\Omega G(x, y)f(y)dy$ for homogeneous boundary data.

Solution. Apply Green's identity to u and $G(x, \cdot)$, use the point-source equation for G, and use the boundary condition to eliminate the boundary term. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 25.17 (Symmetry for self-adjoint problems proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: Under suitable self-adjoint boundary conditions, $G(x, y) = G(y, x)$.

Solution. Apply Green's identity to the two Green functions with poles at x and y, excise small neighborhoods of the poles, and pass to the limit; boundary terms vanish by the shared boundary condition. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 25.18 (Derivative jump in one dimension proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For $-u'' = f$ on an interval, the Green kernel is continuous across $x = y$ and its first derivative has the jump required to create a Dirac mass.

Solution. Integrating the equation $-G_{xx} = \delta_y$ across a small interval around y gives $-(G_x(y^+) - G_x(y^-)) = 1$, while absence of a delta-prime term forces continuity. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 25.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Green functions adapt fundamental solutions to domains and boundary conditions. They separate the singular response to a point source from a correction chosen to enforce the boundary data. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. ☐

25.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 25.20. State and apply the central principle behind Green function definition. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. For a domain operator, $G(x, y)$ solves the differential equation in x with a point source at y and prescribed boundary behavior. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 25.21. State and apply the central principle behind Singular plus regular decomposition. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Typically $G(x, y) = E(x - y) - H(x, y)$, where E is a whole-space fundamental solution and H corrects the boundary values. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 25.22. State and apply the central principle behind Representation formula. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Solutions are expressed by integrating the forcing against G, plus boundary terms when nonhomogeneous data are present. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 25.23. State and apply the central principle behind Symmetry. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Self-adjoint elliptic problems often yield $G(x, y) = G(y, x)$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 25.24. State and apply the central principle behind Positivity. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Maximum principles can imply nonnegativity of Dirichlet Green functions for positive elliptic operators. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 25.25. State and apply the central principle behind One-dimensional construction. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. For second-order ODEs, Green kernels are assembled from two homogeneous solutions matched continuously with a derivative jump. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 25.26. State and apply the central principle behind Boundary dependence. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. Changing Dirichlet to Neumann conditions changes the correction and may require compatibility conditions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Exercise 25.27. State and apply the central principle behind Operator inverse viewpoint. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. ☐

Solution. The Green kernel is the integral kernel of the inverse or resolvent of the boundary-value operator. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 26

Sturm–Liouville Green Kernels

Sturm–Liouville Green kernels combine boundary-value ODE theory, self-adjointness, Wronskians, and spectral expansions. They are the one-dimensional prototype of resolvent kernels for elliptic operators.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

26.1 Sturm–Liouville operator

Consider $Ly = -(py')' + qy$ with separated boundary conditions and positive weight.

26.2 Homogeneous fundamental solutions

Choose one solution satisfying the left boundary condition and one satisfying the right boundary condition.

26.3 Wronskian structure

A weighted Wronskian is constant for homogeneous solutions and controls the denominator of the Green kernel.

26.4 Piecewise kernel

Use one product of homogeneous solutions for $x < \xi$ and the reversed product for $x > \xi$.

26.5 Matching conditions

Continuity at the source and a derivative jump enforce the Dirac equation.

26.6 Self-adjoint symmetry

Self-adjoint boundary conditions make the Green kernel symmetric.

26.7 Eigenfunction expansion

Away from eigenvalues, the resolvent kernel can be expanded in normalized eigenfunctions with spectral denominators.

26.8 Resonance

At an eigenvalue the inverse fails and solvability requires orthogonality compatibility with the nullspace.

26.9 Core structural results

Theorem 26.1 (Green-kernel construction)**.** *Let u satisfy the left boundary condition and v the right one. Away from eigenvalues, $G(x, \xi)$ is proportional to $u(x_<)v(x_>)$ divided by the weighted Wronskian.*

Proof. Each branch solves the homogeneous equation and the boundary condition on its side. Continuity and the prescribed flux jump at $x = \xi$ determine the common normalization by the Wronskian. ☐

Theorem 26.2 (Symmetry)**.** *For a regular self-adjoint Sturm–Liouville problem, $G(x, \xi) = G(\xi, x)$.*

Proof. The product formula is symmetric after swapping x and ξ, or equivalently follows from self-adjointness of the inverse. ☐

Theorem 26.3 (Spectral resolvent expansion)**.** *If $\{\phi_n\}$ is an orthonormal eigenbasis and λ avoids the spectrum, then the Green kernel has formal expansion $G_\lambda(x, \xi) = \sum_n \phi_n(x)\phi_n(\xi)/(\lambda_n - \lambda)$.*

Proof. Expand the forcing in the eigenbasis, solve each scalar coefficient equation, and read off the integral kernel of the resulting inverse. ☐

26.10 Worked examples

Example 26.4 (Sturm–Liouville operator diagnostic)**.** Give a rigorous worked analysis of Sturm–Liouville operator in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Consider $Ly = -(py')' + qy$ with separated boundary conditions and positive weight. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 26.5 (Homogeneous fundamental solutions diagnostic)**.** Give a rigorous worked analysis of Homogeneous fundamental solutions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Choose one solution satisfying the left boundary condition and one satisfying the right boundary condition. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 26.6 (Wronskian structure diagnostic)**.** Give a rigorous worked analysis of Wronskian structure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A weighted Wronskian is constant for homogeneous solutions and controls the denominator of the Green kernel. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 26.7 (Piecewise kernel diagnostic)**.** Give a rigorous worked analysis of Piecewise kernel in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Use one product of homogeneous solutions for $x < \xi$ and the reversed product for $x > \xi$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

26.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 26.8 (Sturm–Liouville operator diagnostic)**.** Give a rigorous worked analysis of Sturm–Liouville operator in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Consider $Ly = -(py')' + qy$ with separated boundary conditions and positive weight. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 26.9 (Homogeneous fundamental solutions diagnostic)**.** Give a rigorous worked analysis of Homogeneous fundamental solutions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Choose one solution satisfying the left boundary condition and one satisfying the right boundary condition. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 26.10 (Wronskian structure diagnostic)**.** Give a rigorous worked analysis of Wronskian structure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A weighted Wronskian is constant for homogeneous solutions and controls the denominator of the Green kernel. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 26.11 (Piecewise kernel diagnostic)**.** Give a rigorous worked analysis of Piecewise kernel in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Use one product of homogeneous solutions for $x < \xi$ and the reversed product for $x > \xi$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 26.12 (Matching conditions diagnostic)**.** Give a rigorous worked analysis of Matching conditions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Continuity at the source and a derivative jump enforce the Dirac equation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 26.13 (Self-adjoint symmetry diagnostic)**.** Give a rigorous worked analysis of Self-adjoint symmetry in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Self-adjoint boundary conditions make the Green kernel symmetric. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 26.14 (Eigenfunction expansion diagnostic)**.** Give a rigorous worked analysis of Eigenfunction expansion in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Away from eigenvalues, the resolvent kernel can be expanded in normalized eigenfunctions with spectral denominators. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 26.15 (Resonance diagnostic)**.** Give a rigorous worked analysis of Resonance in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. At an eigenvalue the inverse fails and solvability requires orthogonality compatibility with the nullspace. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 26.16 (Green-kernel construction proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: Let u satisfy the left boundary condition and v the right one. Away from eigenvalues, $G(x, \xi)$ is proportional to $u(x_<)v(x_>)$ divided by the weighted Wronskian.

Solution. Each branch solves the homogeneous equation and the boundary condition on its side. Continuity and the prescribed flux jump at $x = \xi$ determine the common normalization by the Wronskian. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 26.17 (Symmetry proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For a regular self-adjoint Sturm–Liouville problem, $G(x, \xi) = G(\xi, x)$.

Solution. The product formula is symmetric after swapping x and ξ, or equivalently follows from self-adjointness of the inverse. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 26.18 (Spectral resolvent expansion proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $\{\phi_n\}$ is an orthonormal eigenbasis and λ avoids the spectrum, then the Green kernel has formal expansion $G_\lambda(x, \xi) = \sum_n \phi_n(x)\phi_n(\xi)/(\lambda_n - \lambda)$.

Solution. Expand the forcing in the eigenbasis, solve each scalar coefficient equation, and read off the integral kernel of the resulting inverse. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 26.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Sturm–Liouville Green kernels combine boundary-value ODE theory, self-adjointness, Wronskians, and spectral expansions. They are the one-dimensional prototype of resolvent kernels for elliptic operators. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

26.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 26.20. State and apply the central principle behind Sturm–Liouville operator. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Consider $Ly = -(py')' + qy$ with separated boundary conditions and positive weight. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 26.21. State and apply the central principle behind Homogeneous fundamental solutions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Choose one solution satisfying the left boundary condition and one satisfying the right boundary condition. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 26.22. State and apply the central principle behind Wronskian structure. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A weighted Wronskian is constant for homogeneous solutions and controls the denominator of the Green kernel. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 26.23. State and apply the central principle behind Piecewise kernel. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Use one product of homogeneous solutions for $x < \xi$ and the reversed product for $x > \xi$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 26.24. State and apply the central principle behind Matching conditions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Continuity at the source and a derivative jump enforce the Dirac equation. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 26.25. State and apply the central principle behind Self-adjoint symmetry. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Self-adjoint boundary conditions make the Green kernel symmetric. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 26.26. State and apply the central principle behind Eigenfunction expansion. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Away from eigenvalues, the resolvent kernel can be expanded in normalized eigenfunctions with spectral denominators. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 26.27. State and apply the central principle behind Resonance. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. At an eigenvalue the inverse fails and solvability requires orthogonality compatibility with the nullspace. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 27

Elliptic Operators and Maximum Principles

Maximum principles turn the sign of an elliptic operator into strong qualitative control. They yield uniqueness, comparison, boundary estimates, and positivity of Green functions without explicitly solving the PDE.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

27.1 Elliptic operators

Second-order ellipticity means the principal coefficient matrix is uniformly positive definite.

27.2 Weak maximum principle

For suitable operators, a subsolution cannot achieve a strict positive interior maximum beyond its boundary maximum.

27.3 Strong maximum principle

A nonconstant elliptic solution cannot attain an interior extremum under the standard sign hypotheses.

27.4 Comparison principle

If $Lu \leq Lv$ and the boundary ordering agrees, then the interior ordering follows.

27.5 Uniqueness

The maximum principle applied to the difference of two solutions yields uniqueness for Dirichlet problems.

27.6 Hopf boundary lemma

At a nonconstant boundary extremum, the outward normal derivative has a strict sign under suitable smoothness.

27.7 Positivity preservation

Nonnegative data often produce nonnegative solutions.

27.8 Barriers

Explicit super- and subsolutions transfer geometric boundary information into quantitative estimates.

27.9 Core structural results

Theorem 27.1 (Weak maximum principle)**.** *If $Lu \geq 0$ for a uniformly elliptic operator with appropriate lower-order signs, then the maximum of u is controlled by the boundary maximum.*

Proof. At a hypothetical strict interior maximum, the gradient vanishes and the Hessian is negative semidefinite; ellipticity makes the principal second-order contribution have the wrong sign, contradicting the differential inequality after the standard perturbation argument. □

Theorem 27.2 (Comparison principle)**.** *If $Lu \geq Lv$ in Ω and $u \leq v$ on the boundary, then $u \leq v$ in Ω under the maximum-principle hypotheses.*

Proof. Apply the weak maximum principle to $u - v$. □

Theorem 27.3 (Dirichlet uniqueness)**.** *A homogeneous Dirichlet solution of $Lu = 0$ is identically zero under the maximum-principle hypotheses.*

Proof. Apply the maximum principle to u and to $-u$ to show both maximum and minimum are zero. □

27.10 Worked examples

Example 27.4 (Elliptic operators diagnostic)**.** Give a rigorous worked analysis of Elliptic operators in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Second-order ellipticity means the principal coefficient matrix is uniformly positive definite. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 27.5 (Weak maximum principle diagnostic)**.** Give a rigorous worked analysis of Weak maximum principle in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For suitable operators, a subsolution cannot achieve a strict positive interior maximum beyond its boundary maximum. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 27.6 (Strong maximum principle diagnostic)**.** Give a rigorous worked analysis of Strong maximum principle in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A nonconstant elliptic solution cannot attain an interior extremum under the standard sign hypotheses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 27.7 (Comparison principle diagnostic)**.** Give a rigorous worked analysis of Comparison principle in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. If $Lu \leq Lv$ and the boundary ordering agrees, then the interior ordering follows. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

27.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 27.8 (Elliptic operators diagnostic)**.** Give a rigorous worked analysis of Elliptic operators in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Second-order ellipticity means the principal coefficient matrix is uniformly positive definite. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 27.9 (Weak maximum principle diagnostic)**.** Give a rigorous worked analysis of Weak maximum principle in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For suitable operators, a subsolution cannot achieve a strict positive interior maximum beyond its boundary maximum. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 27.10 (Strong maximum principle diagnostic)**.** Give a rigorous worked analysis of Strong maximum principle in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A nonconstant elliptic solution cannot attain an interior extremum under the standard sign hypotheses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 27.11 (Comparison principle diagnostic)**.** Give a rigorous worked analysis of Comparison principle in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. If $Lu \leq Lv$ and the boundary ordering agrees, then the interior ordering follows. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 27.12 (Uniqueness diagnostic)**.** Give a rigorous worked analysis of Uniqueness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The maximum principle applied to the difference of two solutions yields uniqueness for Dirichlet problems. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 27.13 (Hopf boundary lemma diagnostic)**.** Give a rigorous worked analysis of Hopf boundary lemma in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. At a nonconstant boundary extremum, the outward normal derivative has a strict sign under suitable smoothness. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 27.14 (Positivity preservation diagnostic)**.** Give a rigorous worked analysis of Positivity preservation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Nonnegative data often produce nonnegative solutions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 27.15 (Barriers diagnostic)**.** Give a rigorous worked analysis of Barriers in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Explicit super- and subsolutions transfer geometric boundary information into quantitative estimates. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. ☐

Problem 27.16 (Weak maximum principle proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $Lu \geq 0$ for a uniformly elliptic operator with appropriate lower-order signs, then the maximum of u is controlled by the boundary maximum.

Solution. At a hypothetical strict interior maximum, the gradient vanishes and the Hessian is negative semidefinite; ellipticity makes the principal second-order contribution have the wrong sign, contradicting the differential inequality after the standard perturbation argument. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 27.17 (Comparison principle proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $Lu \geq Lv$ in Ω and $u \leq v$ on the boundary, then $u \leq v$ in Ω under the maximum-principle hypotheses.

Solution. Apply the weak maximum principle to $u - v$. This proof also explains why the stated hypothesis is structural rather than cosmetic. ☐

Problem 27.18 (Dirichlet uniqueness proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: A homogeneous Dirichlet solution of $Lu = 0$ is identically zero under the maximum-principle hypotheses.

Solution. Apply the maximum principle to u and to $-u$ to show both maximum and minimum are zero. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 27.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Maximum principles turn the sign of an elliptic operator into strong qualitative control. They yield uniqueness, comparison, boundary estimates, and positivity of Green functions without explicitly solving the PDE. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

27.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 27.20. State and apply the central principle behind Elliptic operators. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Second-order ellipticity means the principal coefficient matrix is uniformly positive definite. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 27.21. State and apply the central principle behind Weak maximum principle. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. For suitable operators, a subsolution cannot achieve a strict positive interior maximum beyond its boundary maximum. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 27.22. State and apply the central principle behind Strong maximum principle. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A nonconstant elliptic solution cannot attain an interior extremum under the standard sign hypotheses. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 27.23. State and apply the central principle behind Comparison principle. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. If $Lu \leq Lv$ and the boundary ordering agrees, then the interior ordering follows. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 27.24. State and apply the central principle behind Uniqueness. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The maximum principle applied to the difference of two solutions yields uniqueness for Dirichlet problems. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 27.25. State and apply the central principle behind Hopf boundary lemma. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. At a nonconstant boundary extremum, the outward normal derivative has a strict sign under suitable smoothness. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 27.26. State and apply the central principle behind Positivity preservation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Nonnegative data often produce nonnegative solutions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 27.27. State and apply the central principle behind Barriers. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Explicit super- and subsolutions transfer geometric boundary information into quantitative estimates. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 28

Spectral and Transform Methods for PDE

Spectral and transform methods solve linear PDEs by diagonalizing the operator. Fourier transforms diagonalize translation-invariant operators, while eigenfunction expansions diagonalize boundary-value operators on bounded domains.

Learning goals

The reader should be able to state the central definitions precisely, prove the structural results, audit the hypotheses in examples, and use the chapter's estimates in later analysis.

Conceptual roadmap

$$\boxed{\text{structure} \longrightarrow \text{estimate} \longrightarrow \text{limit or representation} \longrightarrow \text{application.}}$$

28.1 Symbol of an operator

A constant-coefficient differential operator becomes multiplication by its polynomial symbol under the Fourier transform.

28.2 Elliptic multipliers

Solving $P(D)u = f$ in frequency space amounts to dividing by the symbol where the division is meaningful.

28.3 Heat equation

The heat transform multiplier $e^{-4\pi^2 t|\xi|^2}$ gives smoothing and decay.

28.4 Wave equation

Frequency modes oscillate with phase determined by the dispersion relation.

28.5 Poisson equation

The Fourier multiplier $|\xi|^{-2}$ represents the Newtonian potential, interpreted distributionally at low frequency.

28.6 Eigenfunction expansions

On bounded domains, self-adjoint elliptic operators admit orthogonal eigenfunctions that replace Fourier exponentials.

28.7 Semigroups

Spectral calculus defines e^{-tL} and links PDE evolution to functional calculus.

28.8 Boundary and whole-space synthesis

Transform methods exploit translation symmetry; spectral methods adapt the same diagonalization idea to domains and boundary conditions.

28.9 Core structural results

Theorem 28.1 (Heat transform solution). *For $u_t = \Delta u$ on $\mathbb{R}^n$ with initial data u_0, $\widehat{u}(t,\xi) = e^{-4\pi^2 t|\xi|^2}\widehat{u}_0(\xi)$.*

Proof. Fourier transform converts the PDE into the scalar ODE $\partial_t \widehat{u} = -4\pi^2|\xi|^2\widehat{u}$ for each frequency; solve the ODE with the transformed initial data. □

Theorem 28.2 (Spectral heat expansion). *If $L\phi_k = \lambda_k\phi_k$ on a bounded self-adjoint problem, then $u(t) = \sum_k e^{-t\lambda_k}\langle u_0, \phi_k\rangle\phi_k$ solves $u_t + Lu = 0$.*

Proof. Expand the initial data in the eigenbasis; each coefficient satisfies the scalar ODE $c_k' + \lambda_k c_k = 0$. □

Theorem 28.3 (Resolvent diagonalization). *For λ off the spectrum, $(L-\lambda)^{-1}$ multiplies each eigenfunction coefficient by $(\lambda_k - \lambda)^{-1}$.*

Proof. Apply $L - \lambda$ to the eigenfunction expansion and solve the scalar coefficient equations independently. □

28.10 Worked examples

Example 28.4 (Symbol of an operator diagnostic). Give a rigorous worked analysis of Symbol of an operator in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A constant-coefficient differential operator becomes multiplication by its polynomial symbol under the Fourier transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 28.5 (Elliptic multipliers diagnostic). Give a rigorous worked analysis of Elliptic multipliers in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Solving $P(D)u = f$ in frequency space amounts to dividing by the symbol where the division is meaningful. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 28.6 (Heat equation diagnostic)**.** Give a rigorous worked analysis of Heat equation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The heat transform multiplier $e^{-4\pi^2 t|\xi|^2}$ gives smoothing and decay. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 28.7 (Wave equation diagnostic)**.** Give a rigorous worked analysis of Wave equation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Frequency modes oscillate with phase determined by the dispersion relation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

28.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 28.8 (Symbol of an operator diagnostic)**.** Give a rigorous worked analysis of Symbol of an operator in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A constant-coefficient differential operator becomes multiplication by its polynomial symbol under the Fourier transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 28.9 (Elliptic multipliers diagnostic)**.** Give a rigorous worked analysis of Elliptic multipliers in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Solving $P(D)u = f$ in frequency space amounts to dividing by the symbol where the division is meaningful. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 28.10 (Heat equation diagnostic)**.** Give a rigorous worked analysis of Heat equation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The heat transform multiplier $e^{-4\pi^2 t|\xi|^2}$ gives smoothing and decay. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 28.11 (Wave equation diagnostic)**.** Give a rigorous worked analysis of Wave equation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Frequency modes oscillate with phase determined by the dispersion relation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 28.12 (Poisson equation diagnostic)**.** Give a rigorous worked analysis of Poisson equation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The Fourier multiplier $|\xi|^{-2}$ represents the Newtonian potential, interpreted distributionally at low frequency. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 28.13 (Eigenfunction expansions diagnostic)**.** Give a rigorous worked analysis of Eigenfunction expansions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. On bounded domains, self-adjoint elliptic operators admit orthogonal eigenfunctions that replace Fourier exponentials. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 28.14 (Semigroups diagnostic)**.** Give a rigorous worked analysis of Semigroups in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Spectral calculus defines e^{-tL} and links PDE evolution to functional calculus. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 28.15 (Boundary and whole-space synthesis diagnostic)**.** Give a rigorous worked analysis of Boundary and whole-space synthesis in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Transform methods exploit translation symmetry; spectral methods adapt the same diagonalization idea to domains and boundary conditions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. □

Problem 28.16 (Heat transform solution proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For $u_t = \Delta u$ on $\mathbb{R}^n$ with initial data u_0, $\widehat{u}(t,\xi) = e^{-4\pi^2 t|\xi|^2}\widehat{u}_0(\xi)$.

Solution. Fourier transform converts the PDE into the scalar ODE $\partial_t \widehat{u} = -4\pi^2|\xi|^2\widehat{u}$ for each frequency; solve the ODE with the transformed initial data. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 28.17 (Spectral heat expansion proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: If $L\phi_k = \lambda_k\phi_k$ on a bounded self-adjoint problem, then $u(t) = \sum_k e^{-t\lambda_k}\langle u_0, \phi_k\rangle\phi_k$ solves $u_t + Lu = 0$.

Solution. Expand the initial data in the eigenbasis; each coefficient satisfies the scalar ODE $c_k' + \lambda_k c_k = 0$. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 28.18 (Resolvent diagonalization proof dossier)**.** Prove the following structural statement and identify the step where its main hypothesis is used: For λ off the spectrum, $(L-\lambda)^{-1}$ multiplies each eigenfunction coefficient by $(\lambda_k - \lambda)^{-1}$.

Solution. Apply $L-\lambda$ to the eigenfunction expansion and solve the scalar coefficient equations independently. This proof also explains why the stated hypothesis is structural rather than cosmetic. □

Problem 28.19 (Chapter synthesis)**.** Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Spectral and transform methods solve linear PDEs by diagonalizing the operator. Fourier transforms diagonalize translation-invariant operators, while eigenfunction expansions diagonalize boundary-value operators on bounded domains. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. □

28.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 28.20. State and apply the central principle behind Symbol of an operator. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. A constant-coefficient differential operator becomes multiplication by its polynomial symbol under the Fourier transform. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 28.21. State and apply the central principle behind Elliptic multipliers. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Solving $P(D)u = f$ in frequency space amounts to dividing by the symbol where the division is meaningful. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 28.22. State and apply the central principle behind Heat equation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The heat transform multiplier $e^{-4\pi^2 t|\xi|^2}$ gives smoothing and decay. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 28.23. State and apply the central principle behind Wave equation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Frequency modes oscillate with phase determined by the dispersion relation. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 28.24. State and apply the central principle behind Poisson equation. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. The Fourier multiplier $|\xi|^{-2}$ represents the Newtonian potential, interpreted distributionally at low frequency. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 28.25. State and apply the central principle behind Eigenfunction expansions. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. On bounded domains, self-adjoint elliptic operators admit orthogonal eigenfunctions that replace Fourier exponentials. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 28.26. State and apply the central principle behind Semigroups. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Spectral calculus defines e^{-tL} and links PDE evolution to functional calculus. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 28.27. State and apply the central principle behind Boundary and whole-space synthesis. Give one immediate consequence in the setting of this chapter.

Hint. Start from the precise definition or estimate in the corresponding section. □

Solution. Transform methods exploit translation symmetry; spectral methods adapt the same diagonalization idea to domains and boundary conditions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.